Production Planning and Execution (PP)

This case study explains an integrated production planning and execution process in detail and thus fosters a thorough understanding of each process step and underlying SAP functionality.

Product

S/4HANA 2023
Global Bike

Fiori 3.0

Level

Undergraduate
Graduate
Beginner

Focus

Production Planning and Execution

Authors

Michael Boldau
Bret Wagner
Stefan Weidner

Version

4.3

Last Update

July 2025

MOTIVATION

The data entry requirements in the production planning exercises (PP 1 through PP 6) were minimized because much of the data already existed in the SAP system. This stored data, known as master data, simplifies the processing of business transactions. Examples for this, are material master data, bills of materials, and routings.

In this case study, we will create consumption values for a finished product to plan and process a complete manufacturing cycle.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to successfully work through this case study, it is not necessary to have finished the PP exercises (PP 1 through PP 6). However, it is recommended.

NOTES

This case study uses the Global Bike data set.

SAP University Alliances

© SAP UCC Magdeburg

Process Overview

Learning Objective Understand and perform a manufacturing process cycle. **Time** 150 min

Scenario In order to experience a complete manufacturing process you will take on different roles within the Global Bike Group, e.g., production supervisor, shop floor worker and plant manager. Overall, you will be working in the Materials Management (MM) and the Production Planning and Execution (PP) departments.

Employees involved

Jun Lee (Production Manager)
Hiro Abe (Plant Manager)
Lars Iseler (Shop Floor Worker 2)
Susanne Castro (Goods Receipt Clerk)
Sanjay Datar (Warehous Employee)
Michael Brauer (Shop Floor Worker 4)
Jamie Shamblin (Controller)

In the following process, you will plan the production for three products (namely, Deluxe Touring Bikes in black, silver, and red) and complete a production order for the red Deluxe Touring Bikes with the help of the system.

Before you can enter the product demands for your touring bike product group, changes in the material master record of the bikes need to be maintained. Afterwards, you will create planned independent requirements for the next 12 months for this product group and convert a selected planned order into a production order. To conclude the process, the production is confirmed as complete, the finished goods are received into the warehouse and costs assigned to the production order are analyzed.

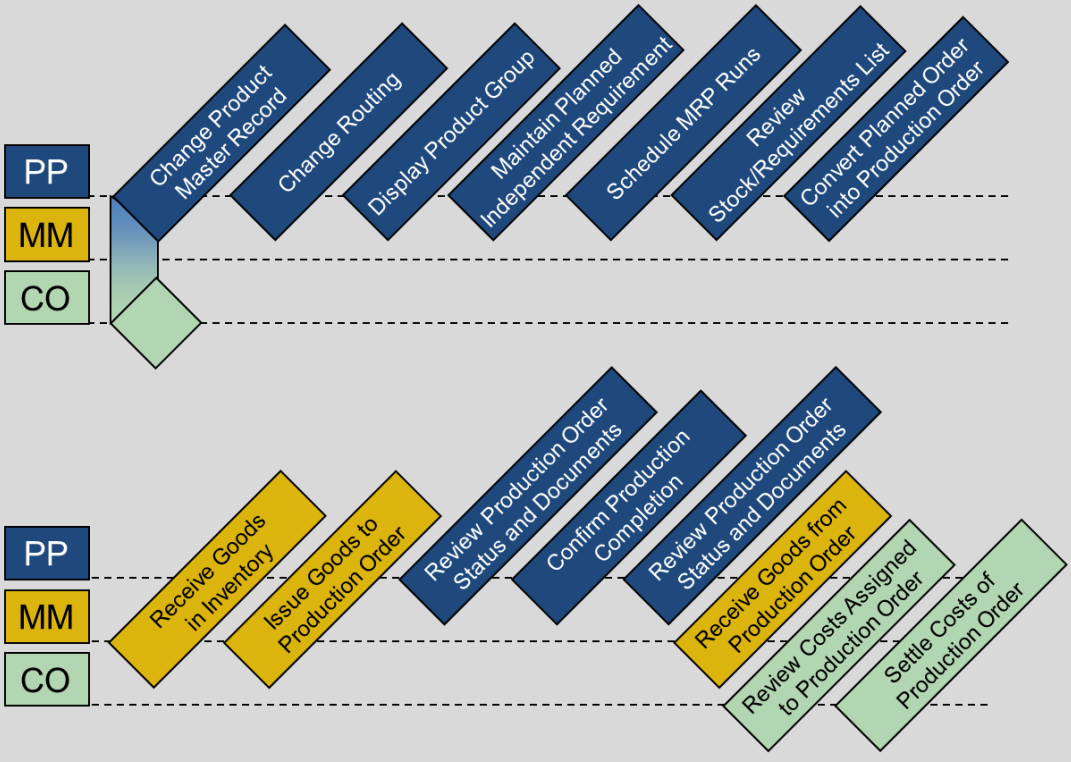


Table of Contents

Process Overview 2
Step 1: Change Product Master Record 4
Step 2: Change Routing 7
Step 3: Display Product Group 10
Step 4: Maintain Planned Independent Requirement 12
Step 5: Schedule MRP Runs 15
Step 6: Review Stock/Requirements List 18
Step 7: Convert Planned Order into Production Order 21
Step 8: Receive Goods in Inventory 23
Step 9: Issue Goods to Production Order 26
Step 10: Review Production Order Status 29
Step 11: Confirm Production Completion 33
Step 12: Review Production Order Status 36
Step 13: Receive Goods from Production Order 38
Step 14: Review Costs Assigned to Production Order 40
Step 15: Settle Costs of Production Order 42
PP Challenge 46

© SAP UCC Magdeburg

Step 1: Change Product Master Record

Task Prepare a product master record for Demand Planning. **Time** 20 min

Short Description In order to plan Global Bike's deluxe touring bikes (black, silver and red) prepare their product master records by adding planning-relevant data to these records.

Name (Position) Jun Lee (Production Manager)

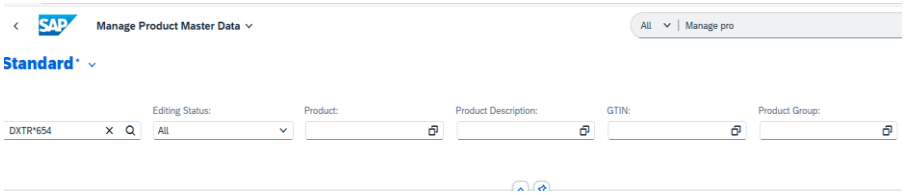
To change the views of a material, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Manage Product Master Data* app.

Start

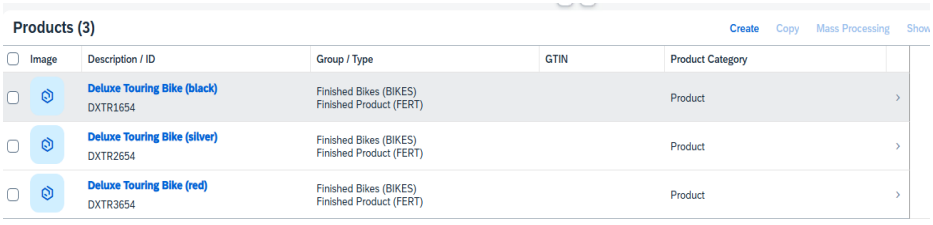


In the search screen, enter **DXTR*###** (replace ### with your three-digit number) in the search field.

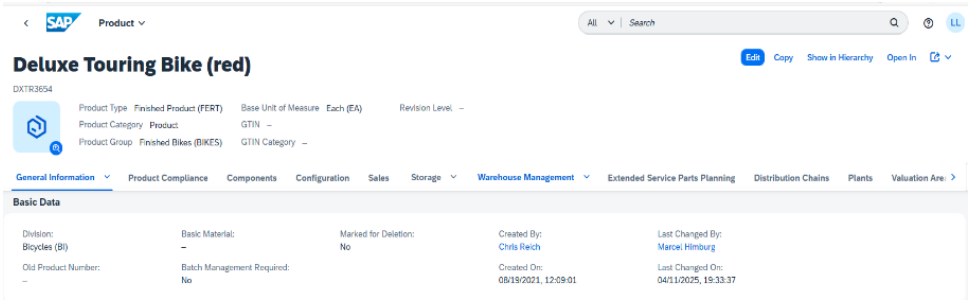
DXTR*###



Press Go. Your various Deluxe Touring Bikes will be displayed.



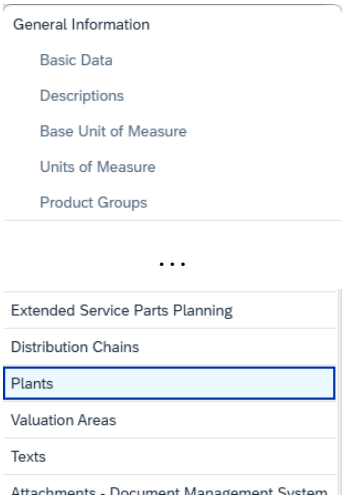
Click the line of **Deluxe Touring Bike (red)** (DXTR3##) to open the details of the product.



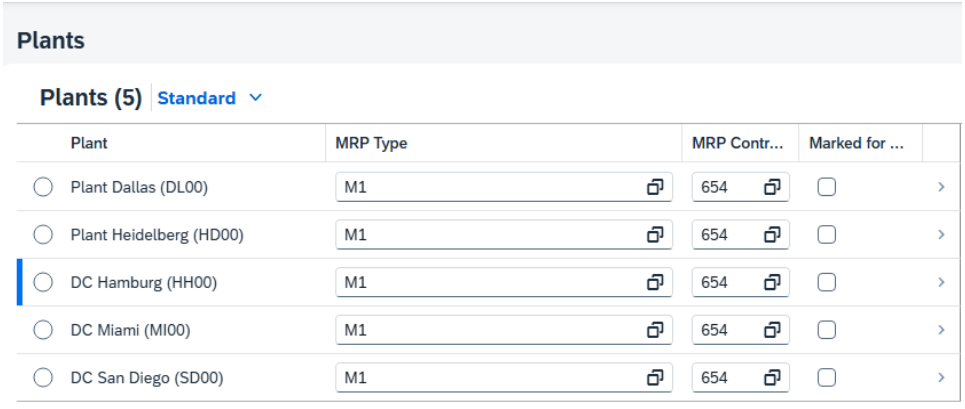
Press **Edit** to switch to the edit mode.

Use the pull-down menu ⌄ to select the *Plants* section. The window automatically scrolls to the correct position.

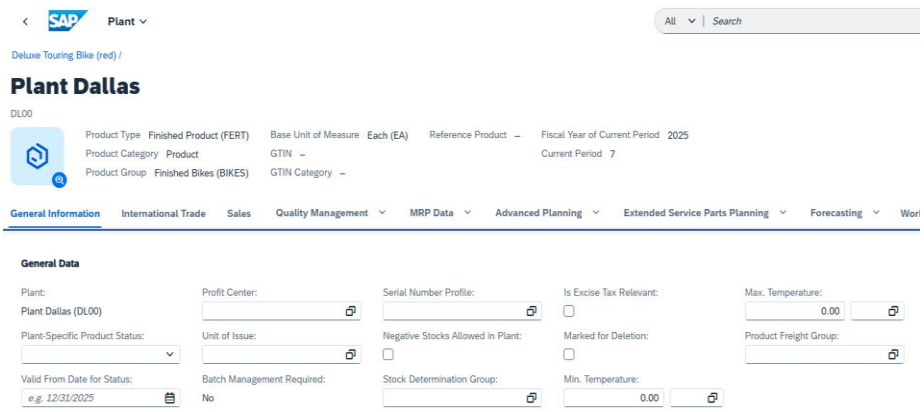
Plants



You will see a list of all plants for which the product has been defined.

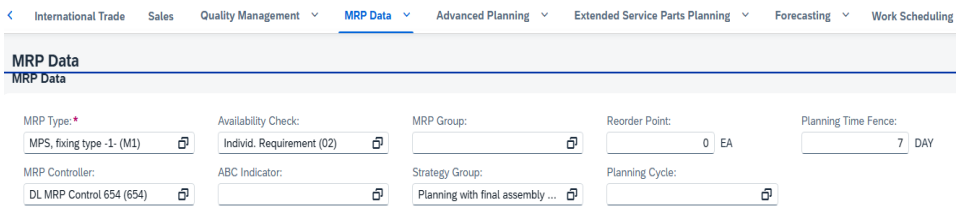


Press › at the end of the line with the DL00 plant to open the plant-specific product master data.



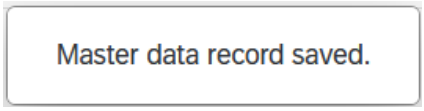
Select the *MRP Data* area. The window automatically scrolls to the correct position. Enter **40** in the *Strategy Group* field (Planning with final assembly).

MRP Data
40



Select **Apply** to save the plant-specific data for plant DL00. Click **Save** to save your changes to the red Deluxe Touring Bike.

The SAP system updates the master data record for material DXTR3### and displays a corresponding message.



Select ‹ to return to the Manage Product Master Data screen.

Repeat the same procedure for the silver and black Deluxe Touring Bike. Start with the silver one (**DXTR2###**) and then change the black bike (**DXTR1###**).

DXTR2###
DXTR1###

Click SAP to return to the SAP Fiori Launchpad.

Step 2: Change Routing

Task Change a routing for a finished good. **Time** 15 min

Short Description Change the routing for your red Deluxe Touring bike.

Name (Position) Jun Lee (Production Manager)

After the operational steps are laid out, the components must be allocated to the individual operations. This is a progressive process where each operation builds off the materials that entered production in the previous operations.

To change the routing, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Manage Routing* app.

Start

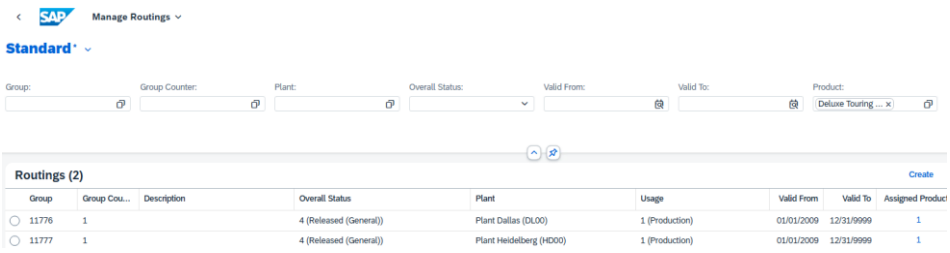


Enter the material number of your red Deluxe Touring bike (**DXTR3###).** In the *Plant* field, enter the Global Bike plant number in Dallas (**DL00**). Then press Go.

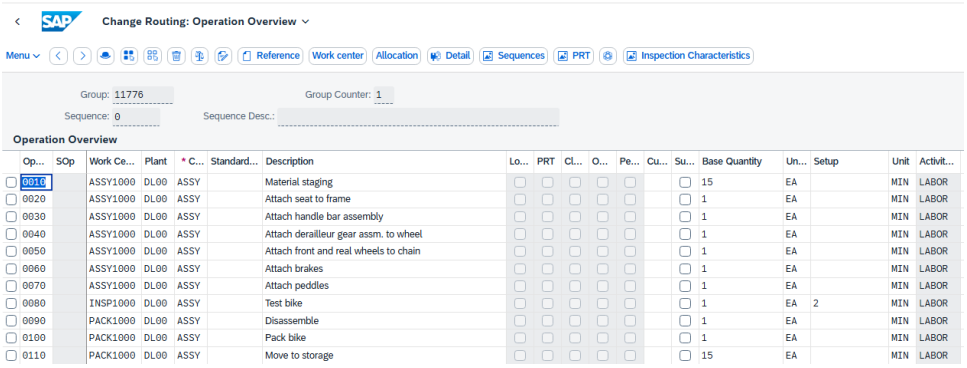
DXTR3###

DL00

Select the displayed routing and then press Change.



The operation overview shows you the individual steps involved in manufacturing the product.



Op...	SOp	Work Ce...	Plant	* C...	Standard...	Description	Lo...	PRT	Cl...	O...	Pe...	Cu...	Su...	Base Quantity	Un...	Setup	Unit	Activit...
0010		ASSY1000	DL00	ASSY		Material staging								15	EA		MIN	LABOR
0020		ASSY1000	DL00	ASSY		Attach seat to frame								1	EA		MIN	LABOR
0030		ASSY1000	DL00	ASSY		Attach handle bar assembly								1	EA		MIN	LABOR
0040		ASSY1000	DL00	ASSY		Attach derailleur gear assm. to wheel								1	EA		MIN	LABOR
0050		ASSY1000	DL00	ASSY		Attach front and real wheels to chain								1	EA		MIN	LABOR
0060		ASSY1000	DL00	ASSY		Attach brakes								1	EA		MIN	LABOR
0070		ASSY1000	DL00	ASSY		Attach peddles								1	EA		MIN	LABOR
0080		INSP1000	DL00	ASSY		Test bike								1	EA	2	MIN	LABOR
0090		PACK1000	DL00	ASSY		Disassemble								1	EA		MIN	LABOR
0100		PACK1000	DL00	ASSY		Pack bike								1	EA		MIN	LABOR
0110		PACK1000	DL00	ASSY		Move to storage								15	EA		MIN	LABOR

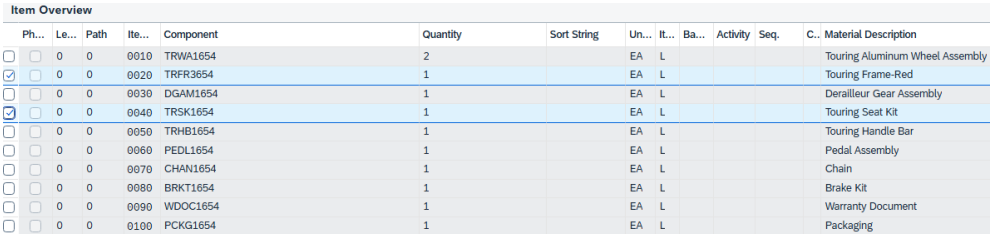
Note The routing group defines a routing and the routing group counter. Moreover, the routing contains a reference to the material whose production is described by the routing.

Besides the standard sequence, it can also have parallel or alternative sequences. Alongside the standard values, the routing also contains time elements that are relevant for scheduling operations. Each operation in the routing may contain its own base quantity, to which these time elements may refer.

Select Allocation to display a list of all components. If this is not displayed, you will find the entry in the pull-down menu under **Menu ► Goto ► Allocation**. Select your material and then click ✓ to display a list of all components.

You now want to assign two components (frame and seat) to the process "Attach seat to frame". Select the rows Touring Frame-Red (**TRFR3##**) and Touring Seat Kit (**TRSK1##**).

TRFR3###
TRSK1###



Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity	Seq.	C..	Material Description
	0	0	0010	TRWA1654	2		EA	L					Touring Aluminum Wheel Assembly
	0	0	0020	TRFR3654	1		EA	L					Touring Frame-Red
	0	0	0030	DGAM1654	1		EA	L					Derailleur Gear Assembly
	0	0	0040	TRSK1654	1		EA	L					Touring Seat Kit
	0	0	0050	TRHB1654	1		EA	L					Touring Handle Bar
	0	0	0060	PEDL1654	1		EA	L					Pedal Assembly
	0	0	0070	CHAN1654	1		EA	L					Chain
	0	0	0080	BRKT1654	1		EA	L					Brake Kit
	0	0	0090	WDOC1654	1		EA	L					Warranty Document
	0	0	0100	PCKG1654	1		EA	L					Packaging

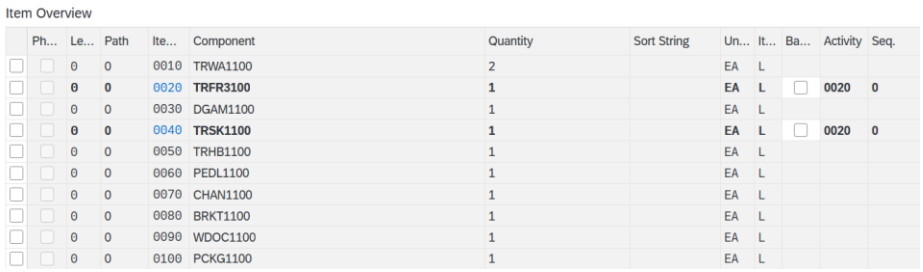
Press New Assignment. In the popup that appears, enter **0020** for *Activity* and confirm the entry with ✓.

0020



Confirm the entry with ✓.

Back in the *Routing Change: Material Component Overview* you can see that both components have now been assigned to *Activity* **0020**.



Item Overview

Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity	Seq.
	0	0	0010	TRWA1100	2		EA	L			
	0	**0**	0020	**TRFR3100**	**1**		**EA**	**L**		**0020**	**0**
	0	0	0030	DGAM1100	1		EA	L			
	0	**0**	0040	**TRSK1100**	**1**		**EA**	**L**		**0020**	**0**
	0	0	0050	TRHB1100	1		EA	L			
	0	0	0060	PEDL1100	1		EA	L			
	0	0	0070	CHAN1100	1		EA	L			
	0	0	0080	BRKT1100	1		EA	L			
	0	0	0090	WDOC1100	1		EA	L			
	0	0	0100	PCKG1100	1		EA	L			

Repeat this process for all other components and assign them to the operations below.

Component	**Operation**
TRHB1### (touring handlebar)	0030
TRWA1### (touring aluminum wheel assembly)	0040
DGAM1### (derailleur gear assembly)	0040
CHAN1### (chain)	0050
BRKT1### (brake kit)	0060
PEDL1### (pedal assembly)	0070
WDOC1### (warranty document)	0100
PCKG1### (packaging)	0100

TRHB1### 0030

TRWA1### 0040

DGAM1### 0040

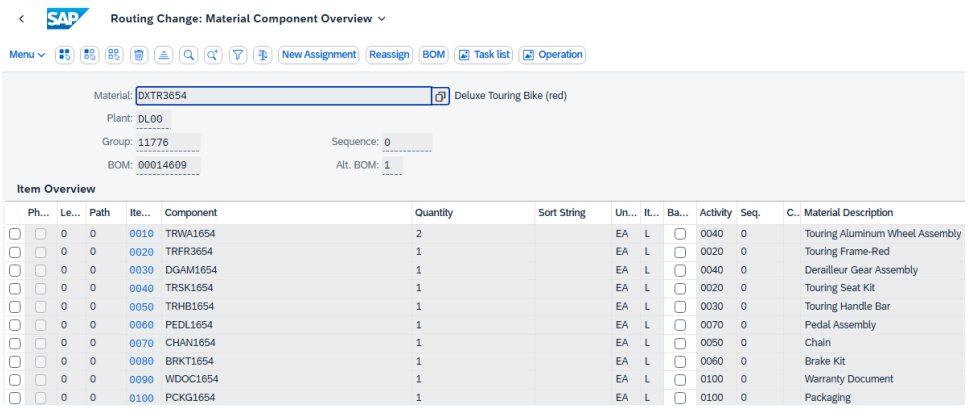
CHAN1### 0050

BRKT1### 0060

PEDL1### 0070

WDOC1### 0100

PCKG1### 0100



Routing Change: Material Component Overview

Menu | New Assignment | Reassign | BOM | Task list | Operation

Material: DXTR3654 Deluxe Touring Bike (red)

Plant: DL00

Group: 11776 Sequence: 0

BOM: 00014609 Alt. BOM: 1

Item Overview

Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity	Seq.	C..	Material Description
	0	0	0010	TRWA1654	2		EA	L		0040	0		Touring Aluminum Wheel Assembly
	0	0	0020	TRFR3654	1		EA	L		0020	0		Touring Frame-Red
	0	0	0030	DGAM1654	1		EA	L		0040	0		Derailleur Gear Assembly
	0	0	0040	TRSK1654	1		EA	L		0020	0		Touring Seat Kit
	0	0	0050	TRHB1654	1		EA	L		0030	0		Touring Handle Bar
	0	0	0060	PEDL1654	1		EA	L		0070	0		Pedal Assembly
	0	0	0070	CHAN1654	1		EA	L		0050	0		Chain
	0	0	0080	BRKT1654	1		EA	L		0060	0		Brake Kit
	0	0	0090	WDOC1654	1		EA	L		0100	0		Warranty Document
	0	0	0100	PCKG1654	1		EA	L		0100	0		Packaging

Apply your changes with Save. The system issues a message that the routing has been saved. Click on ✓ Exit ⓘ .

Click SAP to return to the SAP Fiori Launchpad.

Step 3: Display Product Group

Task Display a product group. **Time** 5 min

Short Description Display the product group (product family) for all your Deluxe Touring bikes.

Name (Position) Jun Lee (Production Manager)

A product group (product family) supports high-level planning. This way, it is not necessary to delve into the minutia of creating planning forecasts for every material in the company.

Product group

To view the Deluxe Touring bike product group, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Display Product Group* app.

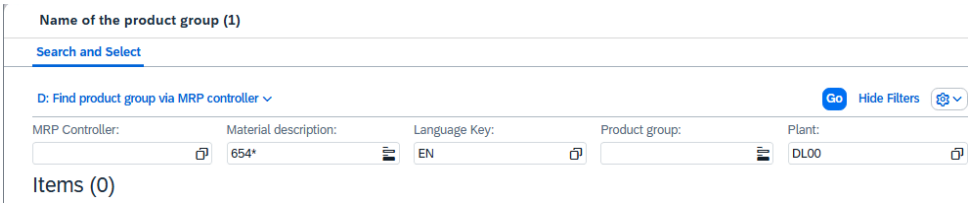
Start



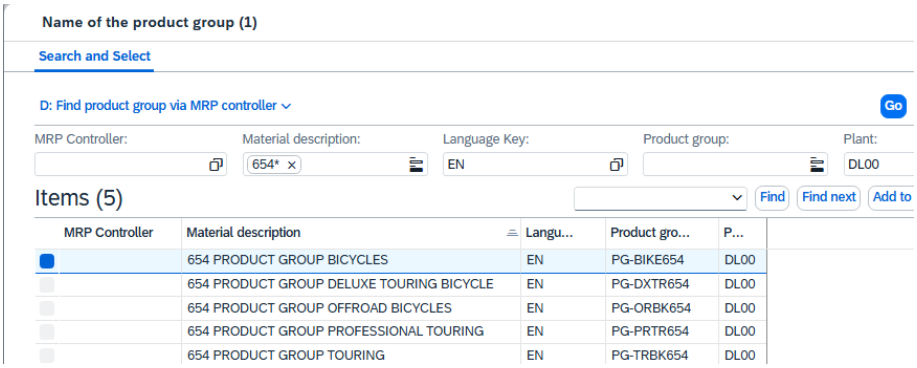
In the *Display Product Group: Initial Screen*, in the *Product group* field find and select your group for deluxe touring bikes. In order to do so, press the search icon (or pressed F4), enter **###*** in the *Material description* field. Remember to replace ### with your three-digit number, e.g., enter 009* if your number is 009. Enter **DL00** as *Plant*.

###*

DL00



Then, press Enter or click on Go to display the search results.

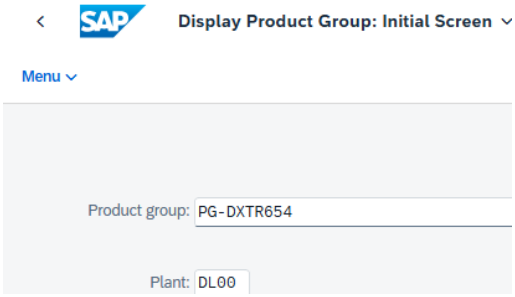


You will see a list of all your product groups, e.g., for mountain bikes or touring bikes.Select the group of Deluxe Touring Bikes (**PG-DXTR##**). Then click OK to apply the selection.

PG-DXTR###

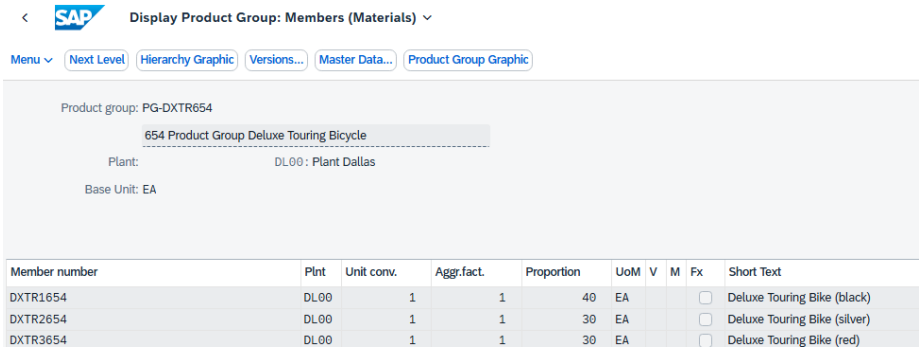
Now that the correct *Product group* (**PG-DXTR###**) is filled in, make sure that *Plant* **DL00** is entered.

DL00



Then, press Enter to display the product group details.

On this screen, you can see that this product group defines proportions for three different bikes: the black, silver and red deluxe touring bike. For the black bike, a share of 40% will be considered and 30% for the silver and the red bikes each.



Click SAP to return to the SAP Fiori Launchpad.

Step 4: Maintain Planned Independent Requirement

Task Create a Planned Independent Requirement for a product group. **Time** 15 min

Short Description Create a 12-month Planned primary requirements for your Deluxe Touring Bike product group.

Name (Position) Jun Lee (Production Manager)

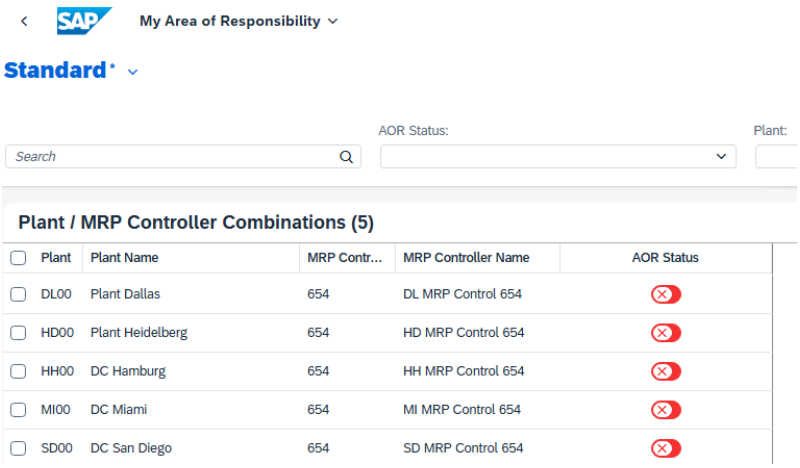
Start

A **planned independent requirement** is a planned requirement notification for a material that is not yet covered by customer orders. It serves as the basis for demand planning in advance planning and consists of a planned quantity at a specific point in time or spread over several periods.

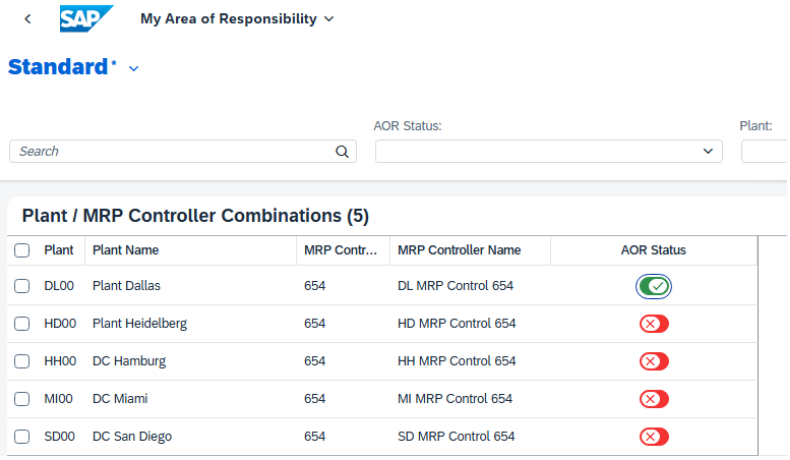
Production Planning and Execution area, in the Production Manager role, use the Maintain Planned Independent Requirements app to create PIRs.



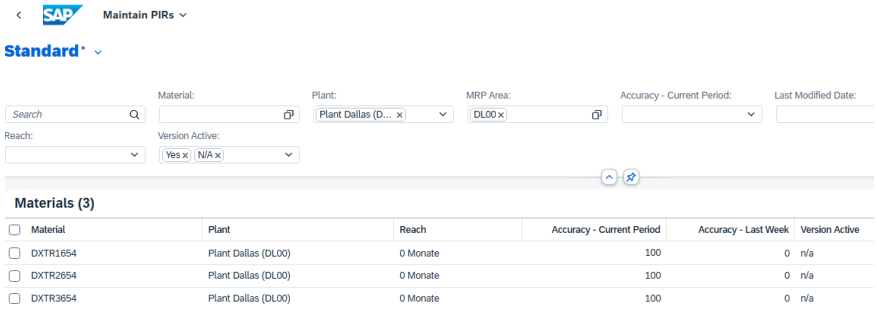
When you first open the app, you'll receive a welcome message asking you to select an area of responsibility. Confirm this with OK, another app, *My Area of Responsibility*, will appear.



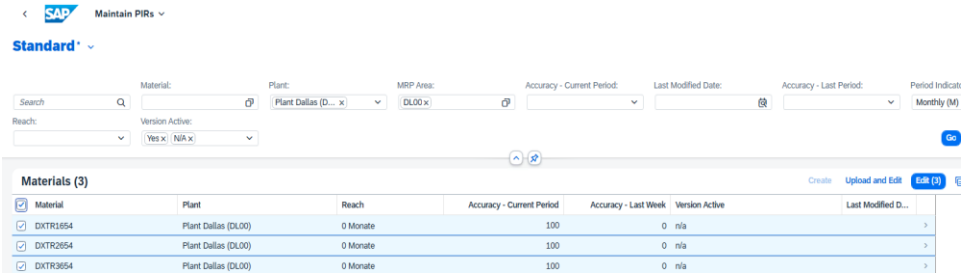
No plant is currently assigned to you. Select the *Dallas plant*



The values are saved automatically here. Go to SAP Fiori Launchpad and open the app *Maintain planned independent requirements* again.

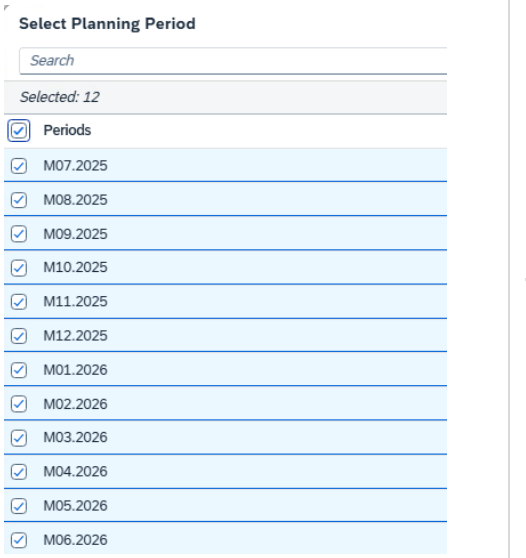


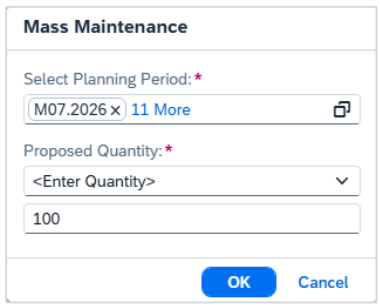
Select all the material and click Edit (3) .



The screen shown is about maintaining Planned Independent Requirements for your materials (*DXTR1###*, *DXTR2###*, *DXTR3###*) in plant *DL00* The requirement values are entered in the table for each period (for the upcoming 12 months). Currently, all values are set to "0". Using the function Mass Maintenance, you can enter the same requirement for multiple periods simultaneously. In the displayed mass maintenance dialog box, **12 periods were** selected (beginning with the current month) and a **suggested requirement of 100 pieces** was entered.

12 periods
100 pieces



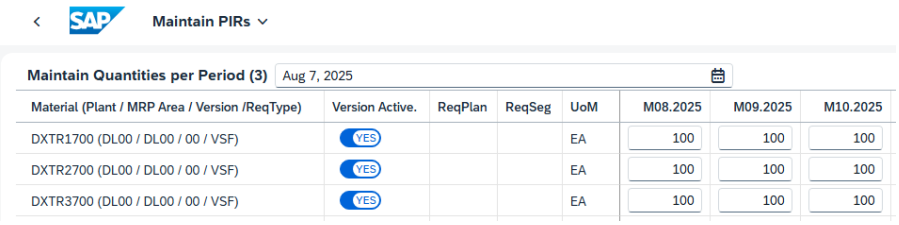


Note

Creation options in mass maintenance: In the Mass Maintenance function in SAP Fiori , you can maintain planned independent requirements in three ways:

- Copy from reference material:
  Adopts the planned independent requirements of another material as a template.
- Enter quantities:
  You manually enter a fixed quantity that will be applied to the selected periods.
- Forecast quantity:
  Uses automatically calculated forecast values from material requirements planning, if available.

By confirming with OK This value is applied to all selected periods at the same time.



Please click on Save.

Click SAP to return to the SAP Fiori Launchpad.

Step 5: Schedule MRP Runs

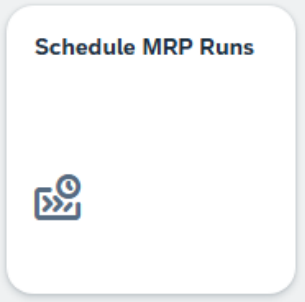
Task Schedule Material Requirements Planning (MRP). **Time** 10 min

Short Description Process a MRP run to generate a series of planned orders that satisfy the requirements from demand management. Concurrently with MPS, the MRP materials will be processed leading to the generation of planned orders for dependent requirements that have been created by the BOM explosion process.

Name (Position) Jun Lee (Production Manager)

To start MRP Scheduling, in the space *Production Planning and Execution* space and the role of *Plant Manager* use the *Schedule MRP Runs* app.

Start

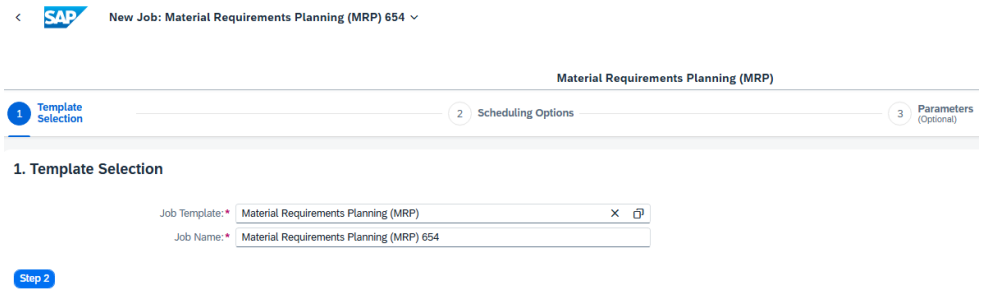


On the Application jobs screen, select Create.

The system then displays the *New Job: Material Requirements Planning (MRP)* screen.

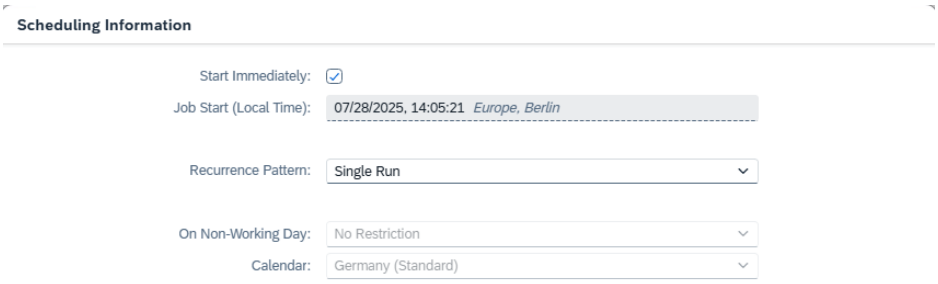
Select the **Material Requirements Planning (MRP)** *job template* and enter a *job name* **Material Requirements Planning (MRP)** ###.

Material Requirements Planning (MRP)
Material Requirements Planning (MRP) ###



Click on Step 2. Define the scheduling options in the *Scheduling options* area. Click on Define Recurrence Pattern and Set *Recurrence pattern* to **Single run**.

Single Run



Click on OK and proceed with Step 3.

Enter as *Plant* **DL00** and product group PG-**DXTR###**. Select the following options in the *Parameters screen area*

DL00
DXTR3###

Section *Also to be included in the planning*

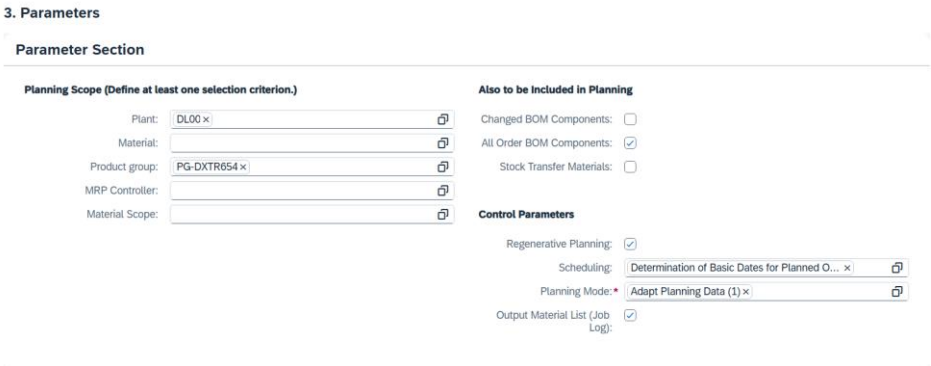
All Order BOM Components

- *All Order BOM Components*: **Select**

Section *Control Parameters*

- Replanning: **Select**
- Scheduling: **1** (Determ. of Basic Dates for Pl. Orders)
- Planning mode: **1** (Adjust Planning Data)
- Output Material List (Job log): S**elect**

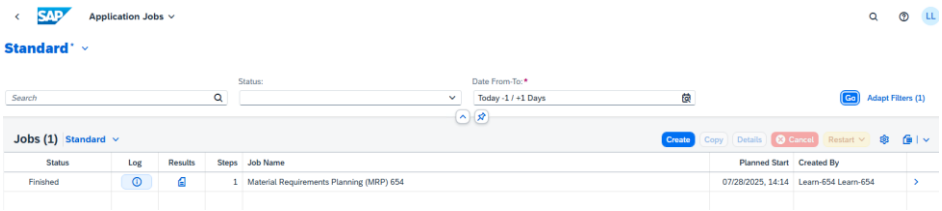
1
1
Output Material List



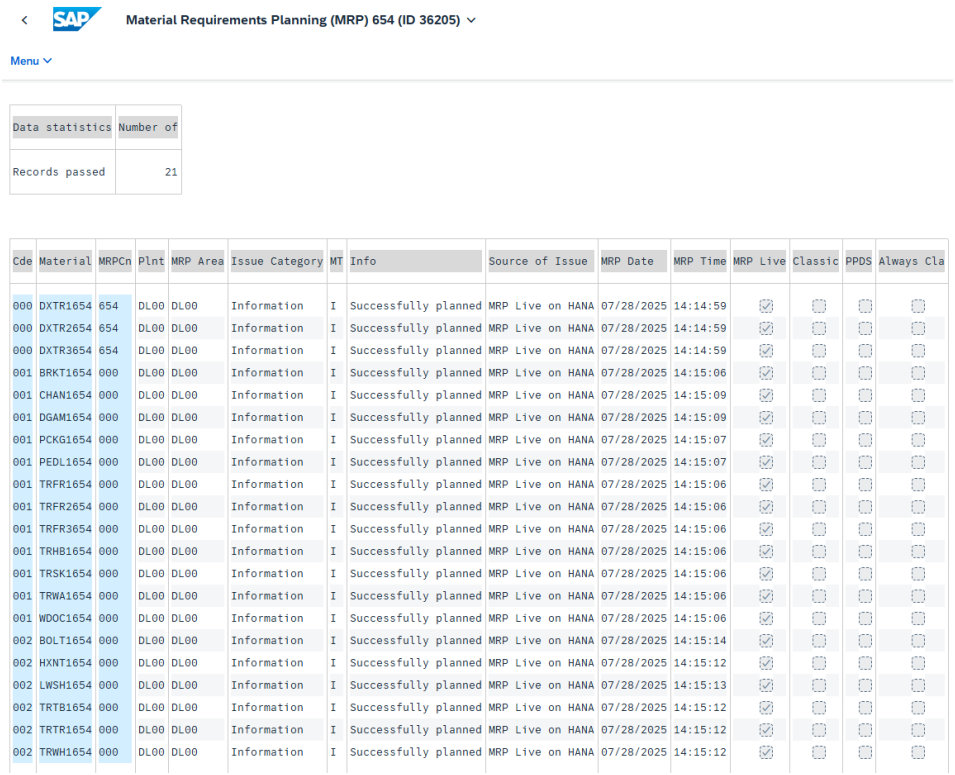
Select Schedule to complete the process.

Note In material requirements planning, a net requirements calculation is carried out to determine whether there is a shortage situation for a material.In addition, the stock and the already existing firm receipts (e.g. purchase orders, production orders, fixed purchase requisitions, and planned orders) are compared with the safety stock and the requirements. This comparison results in the planning-available quantity.

As soon as the planning run is completed, a job is displayed.



Check the planning details of the *results* overview by.



Data statistics	Number of
Records passed	21

Cde	Material	MRPCn	Plnt	MRP Area	Issue Category	MT	Info	Source of Issue	MRP Date	MRP Time	MRP Live	Classic	PPDS	Always Cla
000	DXTR1654	654	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:14:59	☑	☐	☐	☐
000	DXTR2654	654	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:14:59	☑	☐	☐	☐
000	DXTR3654	654	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:14:59	☑	☐	☐	☐
001	BRKT1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
001	CHAN1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:09	☑	☐	☐	☐
001	DGAM1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:09	☑	☐	☐	☐
001	PCKG1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:07	☑	☐	☐	☐
001	PEDL1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:07	☑	☐	☐	☐
001	TRFR1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
001	TRFR2654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
001	TRFR3654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
001	TRHB1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
001	TRSK1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
001	TRWA1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
001	WDOC1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:06	☑	☐	☐	☐
002	BOLT1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:14	☑	☐	☐	☐
002	HXNT1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:12	☑	☐	☐	☐
002	LWSH1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:13	☑	☐	☐	☐
002	TRTB1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:12	☑	☐	☐	☐
002	TRTR1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:12	☑	☐	☐	☐
002	TRWH1654	000	DL00	DL00	Information	I	Successfully planned	MRP Live on HANA	07/28/2025	14:15:12	☑	☐	☐	☐

Click SAP to return to the SAP Fiori Launchpad.

Step 6: Review Stock/Requirements List

Task Review the Stock/Requirements List. **Time** 10 min

Short Description Review the Stock/Requirements List for your deluxe touring bike.

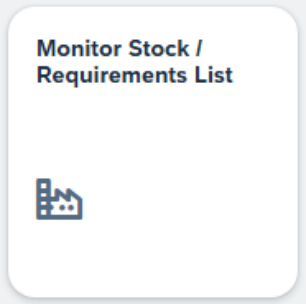
Name (Position) Lars Iseler (Shop Floor Worker 2)

The Stock/Requirements List is a list which dynamically changes whenever a transaction occurs using the given material. Display and review the Stock/Requirements List for all materials of the red deluxe touring bike on hand and the demand that exists against these products. The report shows that there is no stock and therefore nothing is available for use at this time.

Stock/Requirements List

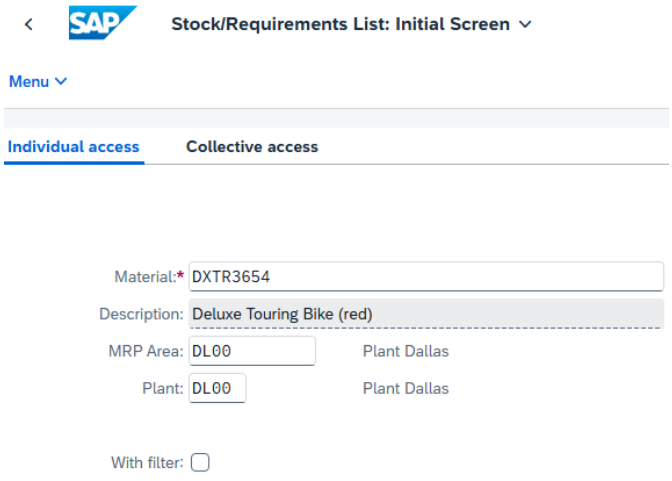
To display the stock/requirements list, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the M*onitor Stock/Requirements List* app.

Start

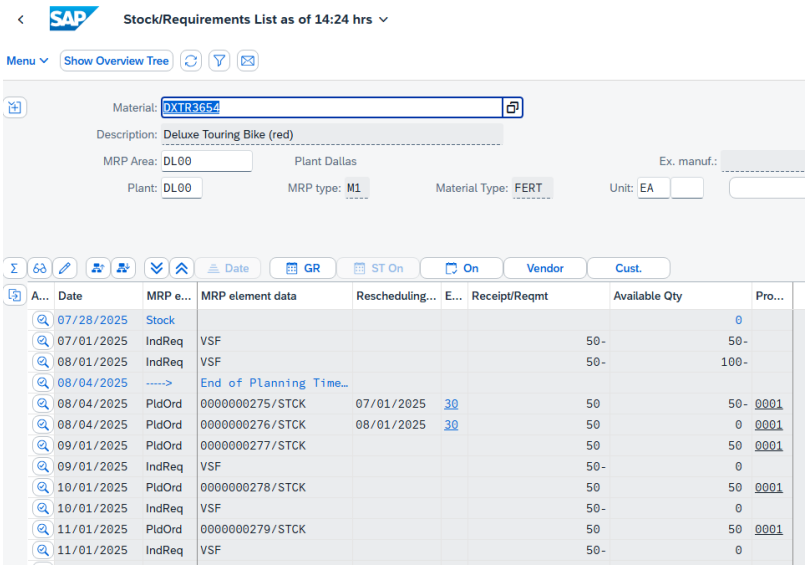


On the *Individual access* tab, enter *Material* **DXTR3###** and *Plant* **DL00**.

DXTR3###
DL00

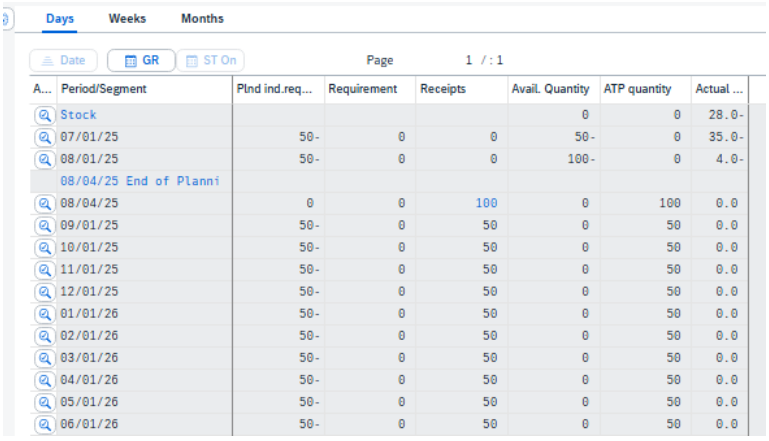


Click on Continue to display the associated stock/requirements list.



A...	Date	MRP e...	MRP element data	Rescheduling...	E...	Receipt/Reqmt	Available Qty	Pro...
	07/28/2025	Stock					0	
	07/01/2025	IndReq	VSF			50-	50-	
	08/01/2025	IndReq	VSF			50-	100-	
	08/04/2025	----->	End of Planning Time...					
	08/04/2025	PldOrd	0000000275/STCK	07/01/2025	30	50	50-	0001
	08/04/2025	PldOrd	0000000276/STCK	08/01/2025	30	50	0	0001
	09/01/2025	PldOrd	0000000277/STCK			50	50	0001
	09/01/2025	IndReq	VSF			50-	0	
	10/01/2025	PldOrd	0000000278/STCK			50	50	0001
	10/01/2025	IndReq	VSF			50-	0	
	11/01/2025	PldOrd	0000000279/STCK			50	50	0001
	11/01/2025	IndReq	VSF			50-	0	

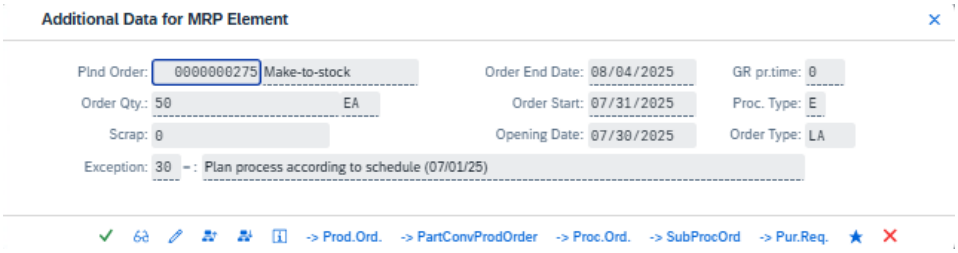
The system currently lists all entries as individual lines. Select Σ to summarize the entries into period totals. This allows you to see the planned independent requirements, planned receipts, and ATP quantities summarized by days, weeks, or months, depending on the selected tab.



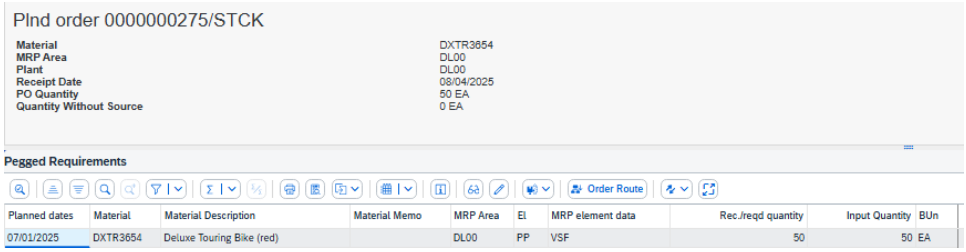
A...	Period/Segment	Plnd ind.req...	Requirement	Receipts	Avail. Quantity	ATP quantity	Actual ...
	Stock				0	0	28.0-
	07/01/25	50-	0	0	50-	0	35.0-
	08/01/25	50-	0	0	100-	0	4.0-
	08/04/25 End of Planni						
	08/04/25	0	0	100	0	100	0.0
	09/01/25	50-	0	50	0	50	0.0
	10/01/25	50-	0	50	0	50	0.0
	11/01/25	50-	0	50	0	50	0.0
	12/01/25	50-	0	50	0	50	0.0
	01/01/26	50-	0	50	0	50	0.0
	02/01/26	50-	0	50	0	50	0.0
	03/01/26	50-	0	50	0	50	0.0
	04/01/26	50-	0	50	0	50	0.0
	05/01/26	50-	0	50	0	50	0.0
	06/01/26	50-	0	50	0	50	0.0

Select to go back to the individual rows.

To view the details of the first planned order (PldOrd), select (Element Details).



Select to display the pegged requirements.



You can see that this planned order is to fulfill our Safety Stock and the first planned independent requirement that was created when we disaggregated our SOP.

Click SAP to return to the SAP Fiori Launchpad.

Step 7: Convert Planned Order into Production Order

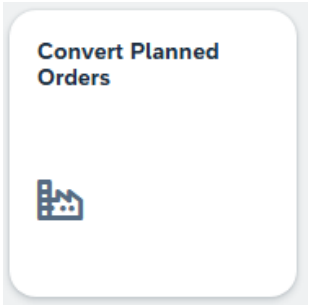
Task Convert a planned order into a production order. **Time** 10 min

Short Description Convert a planned order generated in the MPS/MRP run to a production order. The stock requirements list displays the suggested planned orders from the MPS run.

Name (Position) Lars Iseler (Shop Floor Worker 2)

To convert planned orders into production orders, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the SAP Fiori app *Convert Planned Orders*.

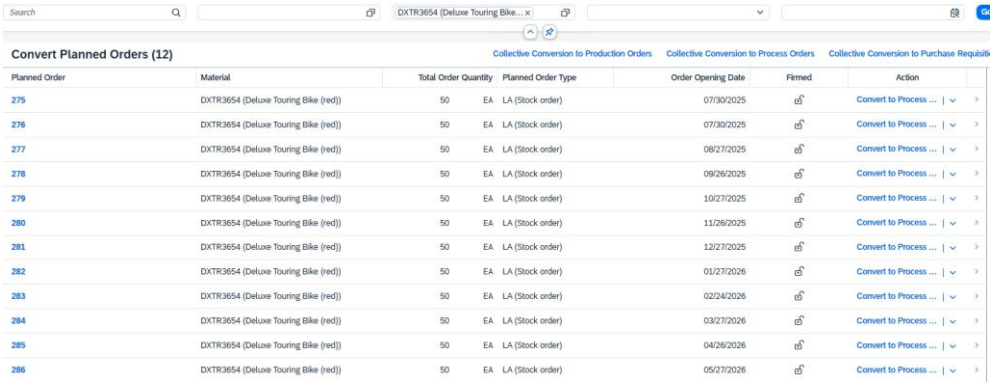
Start



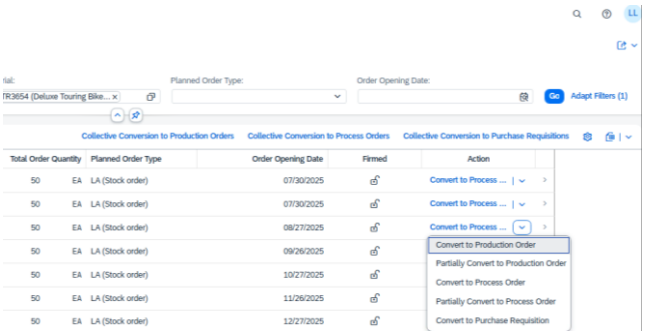
Note You may be asked to define your area of responsibility (cf. Step 4).

Enter *Material* **DXTR3###** and click on Go.

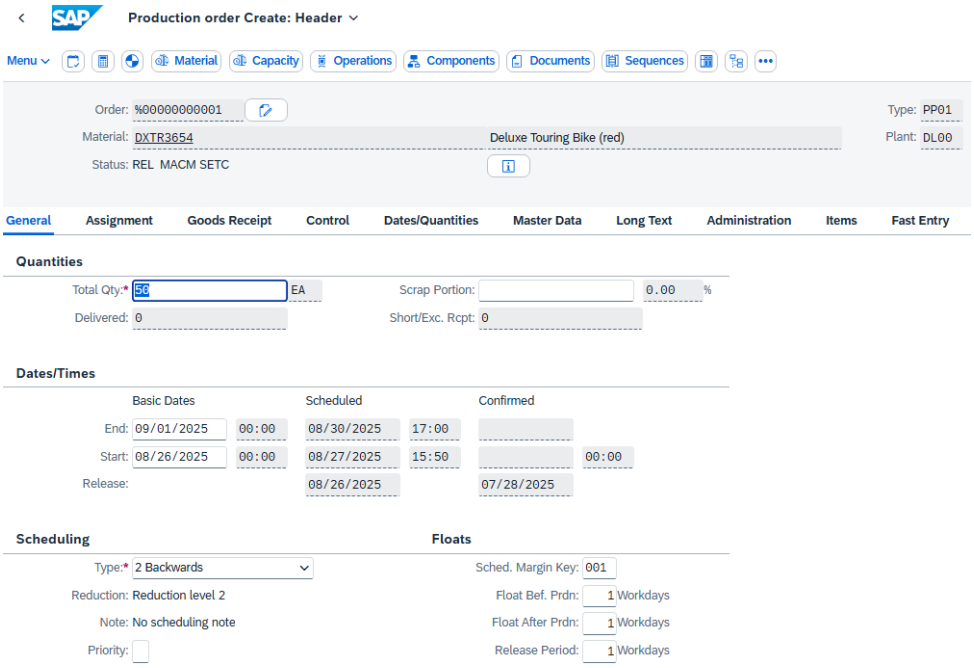
DXTR3###



You will see your planned order. Select the third planned order. In the *Action* tab, you can convert your planned order into a production order using the corresponding button Convert to Production Order.



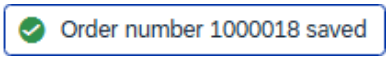
You will be redirected to the *Production Order Create: Header* screen. The order is release automatically.



Click on Save.

Note As soon as you save the production order, the system automatically calculates the planned production costs. This is also displayed accordingly in the status bar.

The system assigns a unique number to the production order. Please make a note of the production order number.



Production order number

Click SAP to return to the SAP Fiori Launchpad.

Step 8: Receive Goods in Inventory

Task Receive goods in the Dallas plant. **Time** 10 min

Short Description Receive enough goods in the Dallas storage locations to start the production process.

Name (Position) Susanne Castro (Goods Receipt Clerk)

Usually, at this point the purchasing department in Dallas would take over and procure enough raw materials from vendors to fill the inventory so that the production process can be initiated. In this case study, we are bypassing this procurement process (this process is explained in the MM unit in detail).

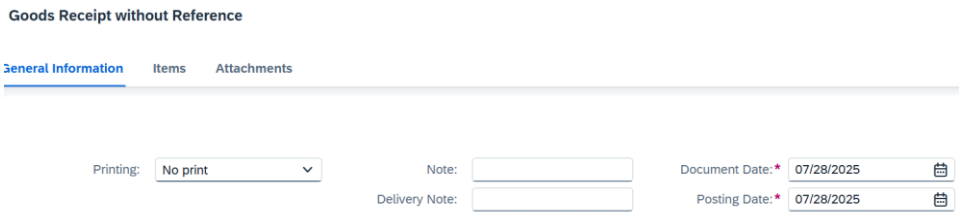
Because the inventory for all DXTR3### components is empty, we will assume that we find 500 pieces each in the storage location.

To receive goods in the inventory, in the space *Production Planning and Execution* space and the role of *Goods Receipt Clerk* use the app *Post Goods Receipt without Reference*.

Start

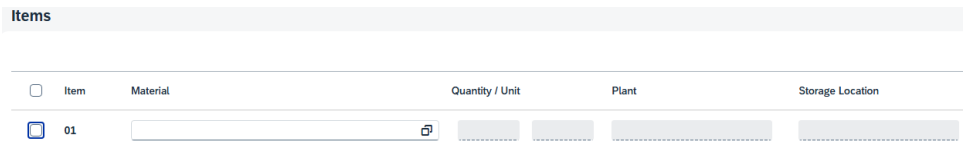


The document and posting date are already defaulted with the current date and can thus be transferred.



Directly below you will find the *Item* section. The table there is ready for input and offers **item 01** in advance.

Items

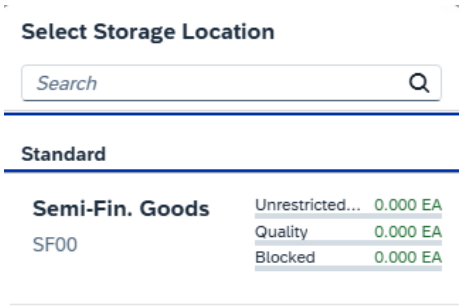


Click on the line of **item 01**, you will switch to a separate input window.

Enter your *Material* **TRWA1###** and press Enter. Now you can enter the *Quantity* **500** with *Unit* **EA**. Next, choose *Plant* **DL00**. When choosing the *Storage Location* your storage for semi-finished goods (**SF00**) is will be proposed. You can check the status directly in the selection screen.

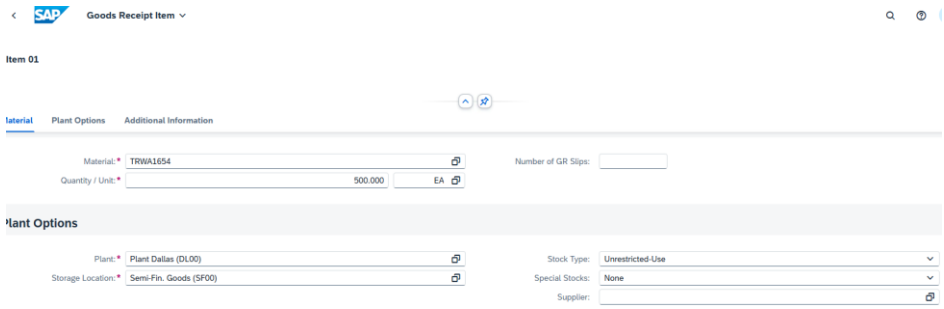
TRWA1###
500 EA

DL00
SF00



Choose the *Storage Location* **SF00**.The *Stock Type* is automatically set to **Unrestricted-Use**, and the *Special Stocks* is **None**.

Unrestricted-Use
None



Click on Apply and New to accept your entries and, at the same time, to be able to specify a new item. The system confirms the transfer of the item.

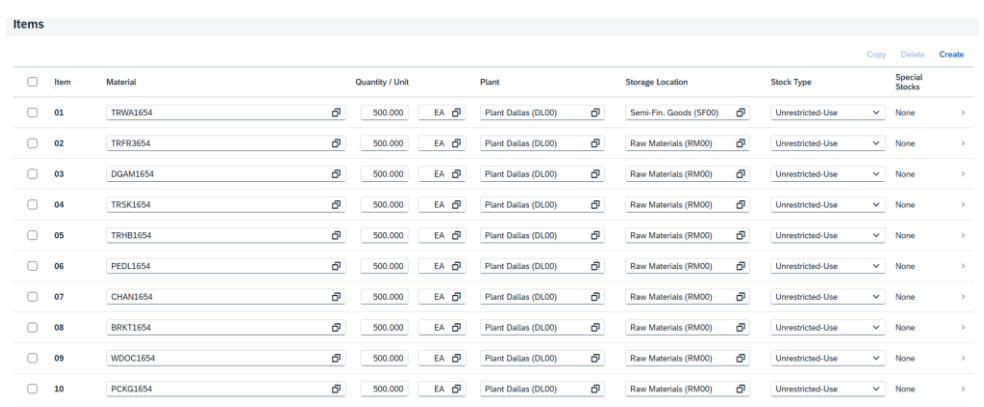
Item 01 is applied.

Now repeat the procedure for the other components of the bike DXTR3###.

Material	Quantity	Unit	Plant	SLoc
TRFR3### (Touring Frame-Red)	500	EA	DL00	RM00
DGAM1### (Derailleur Gear Assembly)	500	EA	DL00	RM00
TRSK1### (Touring Seat Kit)	500	EA	DL00	RM00
TRHB1### (Touring Handle Bar)	500	EA	DL00	RM00
PEDL1### (Pedal Assembly)	500	EA	DL00	RM00
CHAN1### (Chain)	500	EA	DL00	RM00
BRKT1### (Brake Kit)	500	EA	DL00	RM00
WDOC1### (Warranty Document)	500	EA	DL00	RM00
PCKG1### (Packaging)	500	EA	DL00	RM00

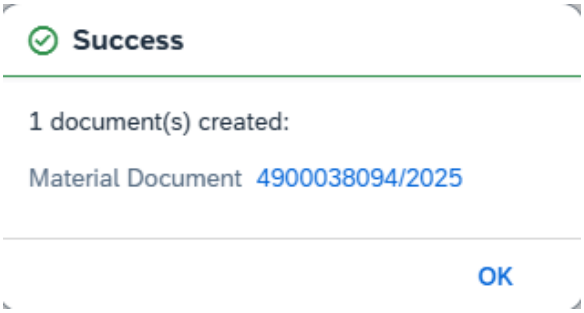
TRFR3###
DGAM1###
TRSK1###
TRHB1###
PEDL1###
CHAN1###
BRKT1###
WDOC1###
PCKG1###

As soon as you create the last item, confirm it with Apply to automatically return to the goods receipt posting. There you will see all created items.



Note If you have forgotten an item, you can add further items by clicking on Create Furthermore, you can also correct entries if necessary.

Save your goods receipt with Post.The SAP system will assign a unique number to the goods receipt and issue an associated message.



Material Document Number

Confirm the success message with OK .

Click SAP to return to the SAP Fiori Launchpad.

Step 9: Issue Goods to Production Order

Task Issue goods to a production order. **Time** 10 min

Short Description Now that all necessary components are on stock issue them to your production order in precise quantity.

Name (Position) Sanjay Datar (Warehouse Employee)

The goods issue process is fully defined in the production order, BOM, and routing. The quantities and the materials are reserved for this specific production order, they will be withdrawn with reference to the order number and will be used to assign actual costs to the production order for managerial accounting purposes.

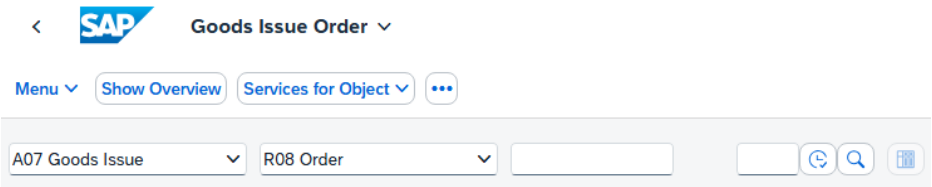
To issue goods to a production order, in the space *Production Planning, Execution space*, and the role of *Warehouse Employee* use the app *Post Goods Movement*.

Start



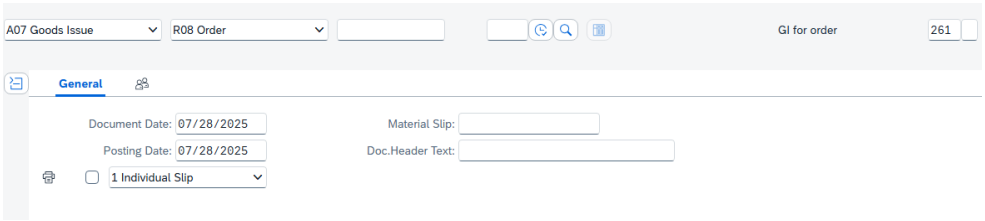
Make sure that **Goods Issue** and **Order** are selected in the drop-down menus.

Goods Issue
Order



The *Document Date* as well as the *Posting Date* should be already set to the current date. The *Movement Type* should be set to **261** (GI for order).

261



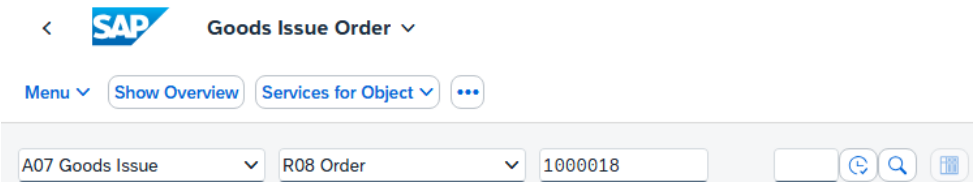
Enter your noted **production order** number.

production order

Process Orders using the Info System DXTR3###

Alternatively, click the value help icon in the order field. In the *Order Number (1)* popup, use the icon on the far right to display a list of all tabs. Select the tab *Production Orders using the Info System*. On this tab, enter your material **DXTR3###** in the *Material* field and click Execute. Select your order and accept it with .

Once you have found or entered your production order number, press to load the order details.



Note Goods issues posting for the required components is another milestone in the production order process. The following functions are performed when a GI for the components of the production order is posted:

- Storage-location-specific update of the stock and consumption fields
- Reduction of the reservation (for planned withdrawal)
- Update of costs for unplanned withdrawals
- Determination of actual costs (valuation) and order update
- Consumption update
- Generation of material and accounting documents (FI and CO documents)
- Creation of material document.
- Creation of accounting document
- Creation of controlling document
- Printing of GI document

The goods issues posting is controlled through a movement type (261), to which each posting refers. This can take place manually or automatically.

SF00
RM00

An itemized list will appear. It lists all the materials and their respective quantities that need to be issued to your order. You need to tell the system what Storage Location the materials should be withdrawn from. For the Touring Aluminum Wheel Assembly (TRWA1###), enter **SF00** (Semi-finished goods) and for all other materials **RM00** (Raw materials) in the SLoc fields. Before pressing Enter, compare your screen with the one shown below. Notice that your quantity could be different.

Note If you are unable to enter the storage location for a particular material, this is due to the open detail view in the lower part of the screen. Once you minimize this view, you can continue as usual.

OK

Flag each item with **OK.** If you cannot flag the first item please close the detail view with a click on unter the line item list.

Line	Mat. Short Text	Wa...	OK	Qty in UnE	EUn	S...	SLoc	Order	Acti...	Bus...	Co...
1	Touring Frame-Red	☐	☑	160	EA		RM00	1000003	0020	BI00	US00
2	Derailleur Gear Assembly	☐	☑	160	EA		RM00	1000003	0040	BI00	US00
3	Touring Seat Kit	☐	☑	160	EA		RM00	1000003	0020	BI00	US00
4	Pedal Assembly	☐	☑	160	EA		RM00	1000003	0070	BI00	US00
5	Touring Handle Bar	☐	☑	160	EA		RM00	1000003	0030	BI00	US00
6	Chain	☐	☑	160	EA		RM00	1000003	0050	BI00	US00
7	Brake Kit	☐	☑	160	EA		RM00	1000003	0060	BI00	US00
8	Warranty Document	☐	☑	160	EA		RM00	1000003	0100	BI00	US00
9	Packaging	☐	☑	160	EA		RM00	1000003	0100	BI00	US00
10	Touring Aluminum Wheel Assembly	☐	☑	320	EA		SF00	1000003	0040	BI00	US00

Click on Post and record the material document number.

Material document 4900038095 posted

Material Document Number

_____

Click SAP to return to the SAP Fiori Launchpad.

Step 10: Review Production Order Status

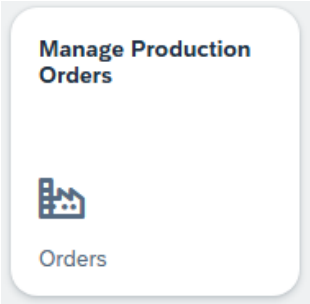
Task Review the production order status. **Time** 10 min

Short Description Review the current production order with respect to the status of the order.

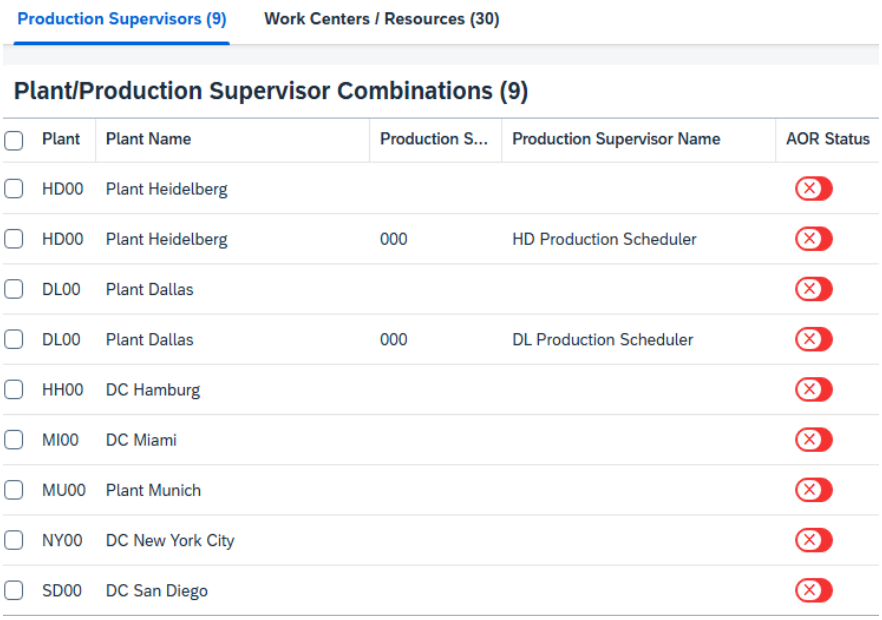
Name (Position) Michael Brauer (Shop Floor Worker 4)

To display the production order, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the app *Manage Production Orders*.

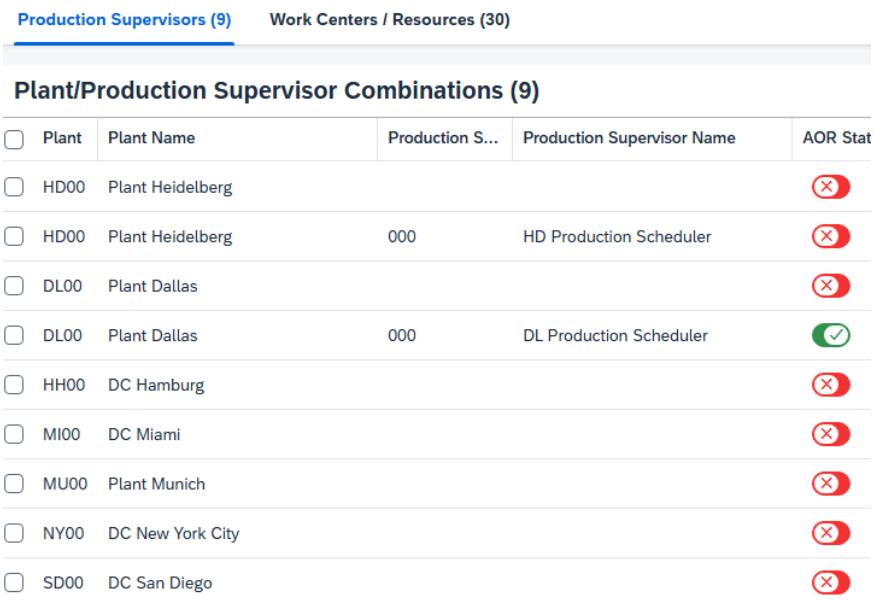
Start



When you open the app for the first time, you will see a welcome message telling you to select an area of responsibility. Confirm this with OK, another popup will appear.

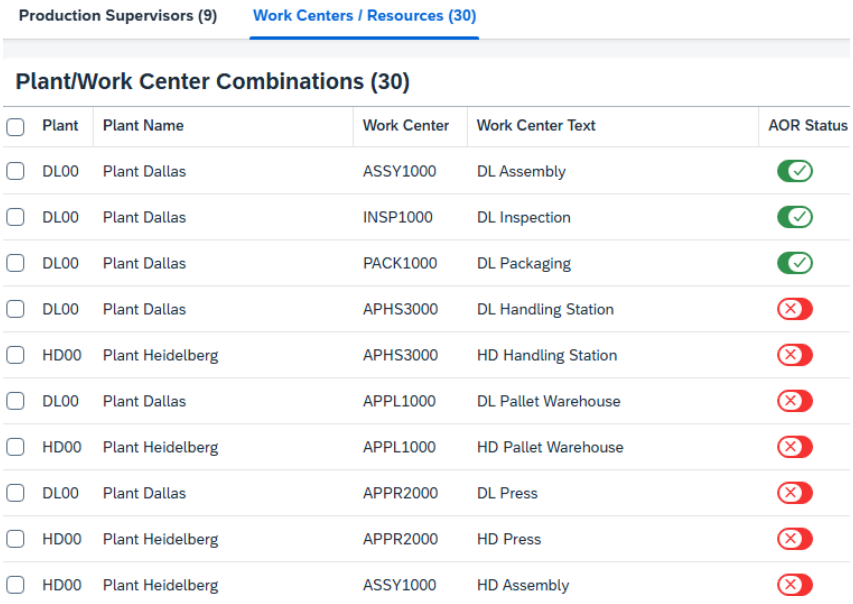


Choose *Production Supervisor*. No Plant is currently assigned. Select the plant in Dallas.

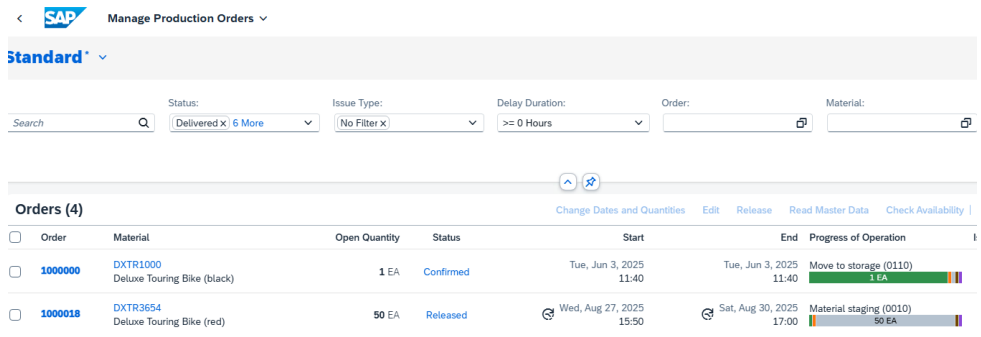


Now select the tab Work Centers/Resources. Choose the *Work Center* **DL Assembly (**ASSY1000**), DL Inspection (**INSP1000**)** and **DL Packaging (**PACK1000**)** from your Pland Dallas

DLAssembly
DL Inspection
DL Packaging

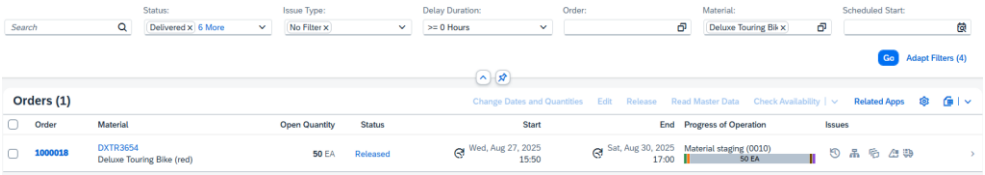


The values are saved automatically here. Go to SAP Fiori Launchpad and open the app *Edit production orders* again. You will receive an overview of all existing orders. Depending on your course progress, there may be multiple production orders with different processing statuses.



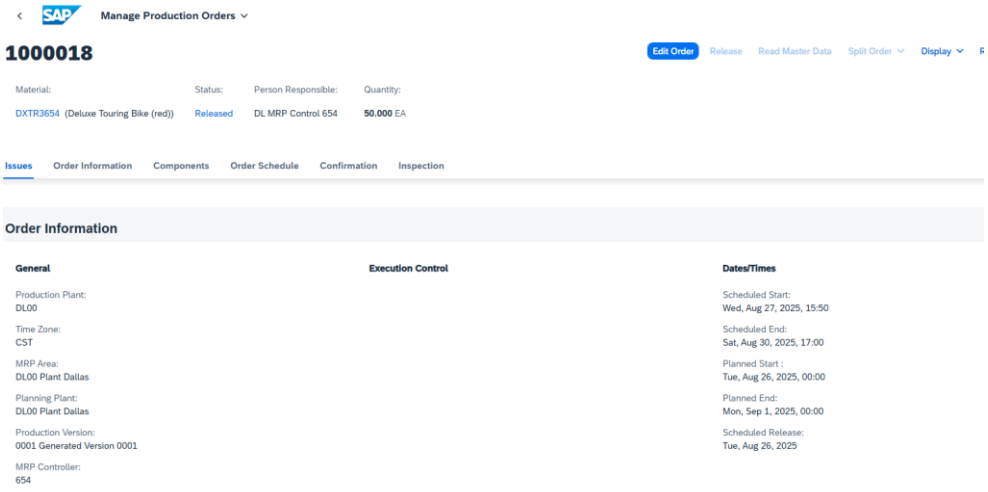
In the field *Material* enter your material **DXTR3###** and choose Go to display your order.

DXTR3###

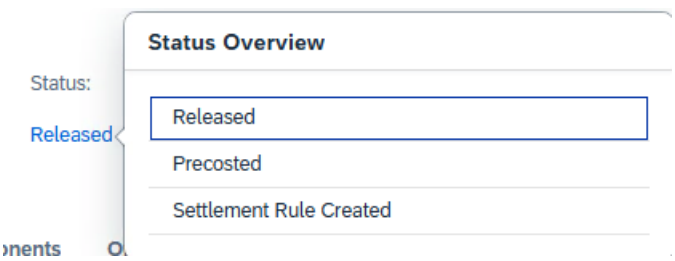


The tabular overview already provides you with various information about your order, such as the current status and the current processing status.

For further information select the entry. You will be forwarded to the details of the production order.

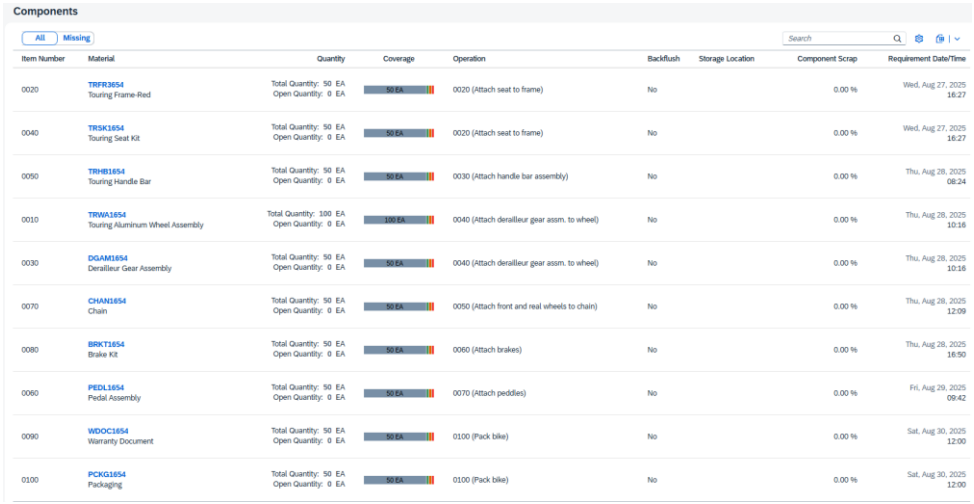


Click on the status Released for more information. You can see that your production order is precosted and a settlement rule has been created.



Components

Now click on the *Components* tab. The screen scrolls to the corresponding position.



In the last task, you posted the goods issue for the production order. In the production order, you now see that there are no more open quantities for this order.

Click SAP to return to the SAP Fiori Launchpad.

Step 11: Confirm Production Completion

Task Confirm production order completion. **Time** 5 min

Short Description Confirm completion for your production order.

Name (Position) Michael Brauer (Shop Floor Worker 4)

When the assembly has been completed for the current production order, we need to confirm that certain procedures and activities have been completed and record the quantity of the finished product that has been manufactured.

Production completion

To confirm production completion, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the app *Enter Production order Confirmation.*

Start

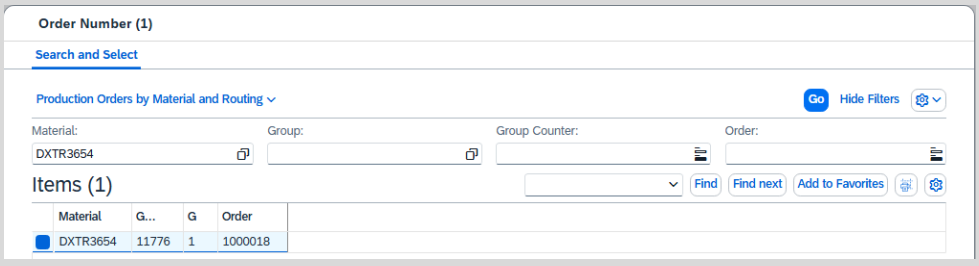


Enter your **production order** number and click on Enter.

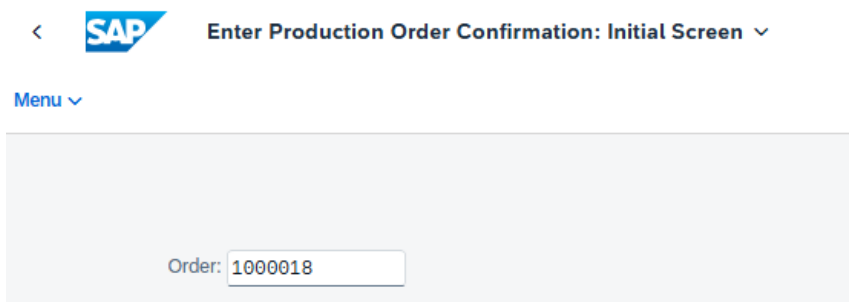
Production order number

Note If you no longer know this, you can alternatively search for your order. To do so, use the value help symbol in the *order field.* . In the popup, go to the search *for production orders for the material and standard routing* and enter your material number **DXTR3** ###. Click on Starten. Select your order and accept it with OK.

DXTR3 ###



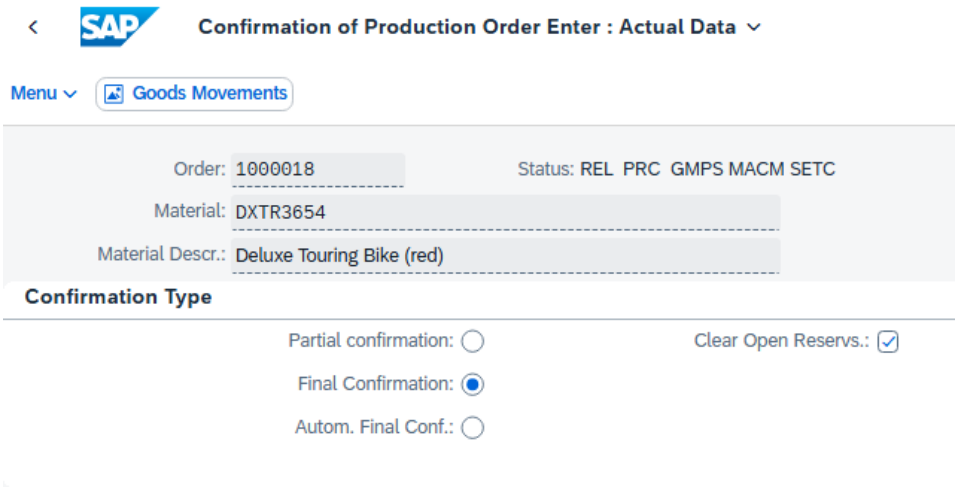
Final Confirm.
Clear Reservation Amount



Then click on Enter

Check if **Final Confirmation** and **Clear Open Reserv.** is already checked in the *Confirmation Type* section.

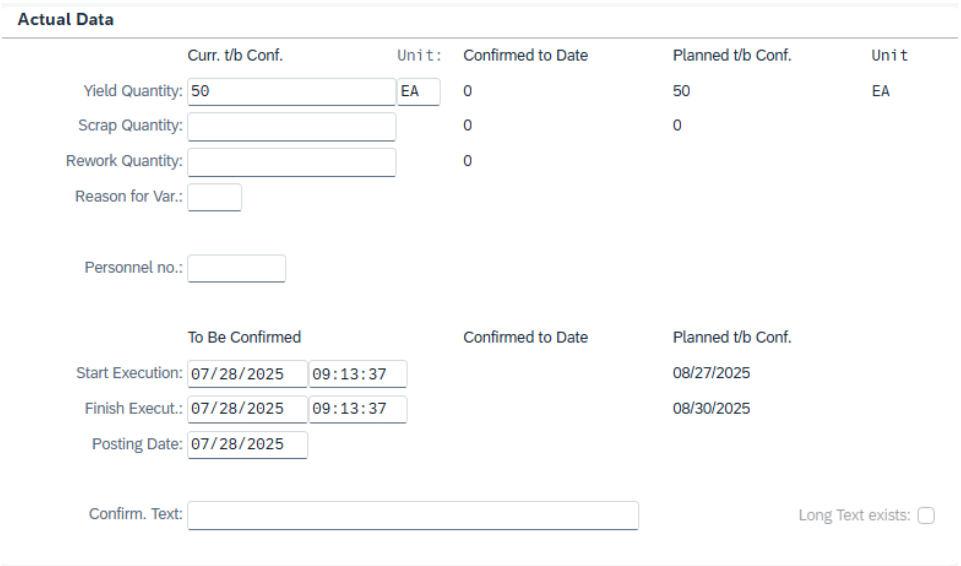
Final Confirmation
Clear Open Reserv



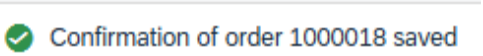
Furthermore, in the *Actual Data* tab, the quantity of bicycles that you should produce for this order should already be entered in the *Yield Quantity* field.

Change the *Start Execution* to **1 hour earlier** than the preset time.

1 hour earlier



Save your entries with Save. You will get a confirmation from the system.



Note When the confirmation is saved, labor costs for the order are calculated automatically. The quantity yield also establishes the parameters for the goods receipt into Inventory.

Click to go to SAP Fiori Return to Launchpad.

Step 12: Review Production Order Status

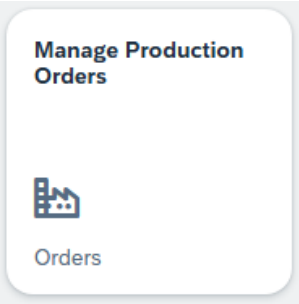
Task Review the production order status. **Zeit** 5 Min.

Short Description Review the current production order with respect to the status of the order.

Name (Position) Michael Brauer (Shop Floor Worker 4)

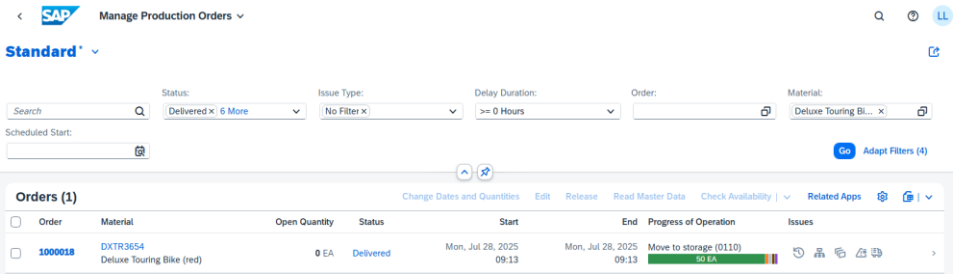
To display the production order, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the app *Manage Production Orders*.

Start



In the field *Material* enter your material **DXTR3###** and choose Go to display your order.

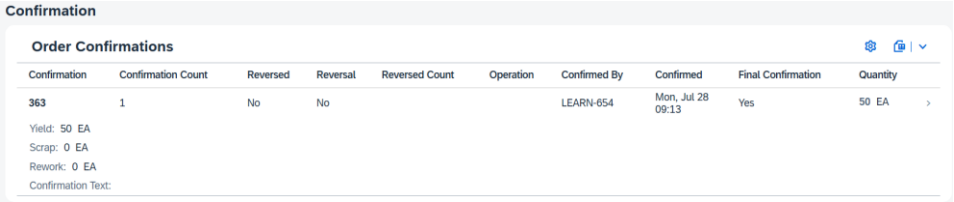
DXTR3###



As you can see, the status of your production order has changed from *Released* to *Confirmed*. Furthermore, the processing status is now at *Move to storage*.

Confirmation

For more information, select the entry, you will be redirected to the details of the production order. Click on the *Confirmation* tab to go to the related area.



An order confirmation is now available. You can see that the complete quantity of your production order has been confirmed and that there is no scrap.

After the confirmation, the goods receipt now needs to take place to complete the order.

Click  to return to the SAP Fiori Launchpad.

Step 13: Receive Goods from Production Order

Task Post a goods receipt from production order. **Time** 15 min

Short Description Post a goods receipt from your production order.

Name (Position) Susanne Castro (Goods Receipt Clerk)

Goods receipt

Receive the completed products into finished goods inventory. Check the quantity proposed against the quantity specified in the production order and the quantity specified during confirmation. If there are any discrepancies, the system will decide if an error or warning message should be generated, depending upon the deviation identified.

Start

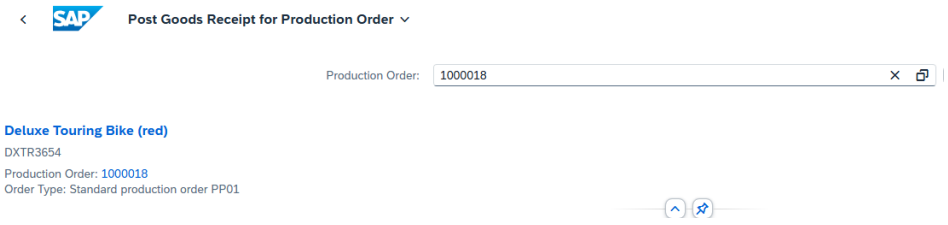
To post the goods receipt, in the space *Production Planning and Execution* space and the role of *Goods Receipt Clerk* use the app *Post goods receipt for production order*.



Production order

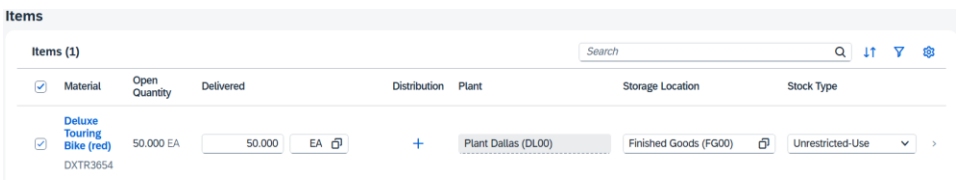
DXTR3###

Enter your noted **production order**. Alternatively, click the input help icon and enter your product **DXTR3###** in the search. Press Start and then select your production order from the results list. After you have entered or found your number, press Enter. Your production order will be loaded and displayed

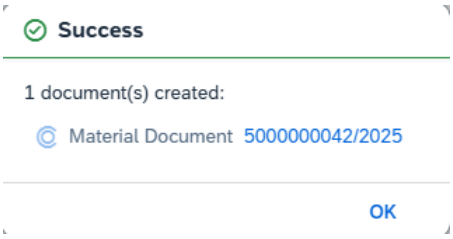


FG00

Add the storage location **FG00** for end products, all other settings can be accepted.



Save your goods receipt with Post. The SAP system will assign a unique number to the goods receipt and issue a corresponding message.



Material Document Number

Furthermore, this updates the current value of the material produced to the production order. Confirm the successmessage with OK.

Click SAP to return to the SAP Fiori Launchpad.

Step 14: Review Costs Assigned to Production Order

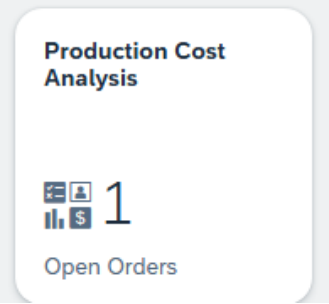
Task View the corresponding costs assigned to your production order. **Zeit** 5 Min.

Description Review all costs associated with your production order.

Name (Position) Jamie Shamblin (Controller)

To view the cost of a manufacturing job, in the space *Production Planning and Execution* space and the role of *Controller* use the *Production Cost Analysis* app.

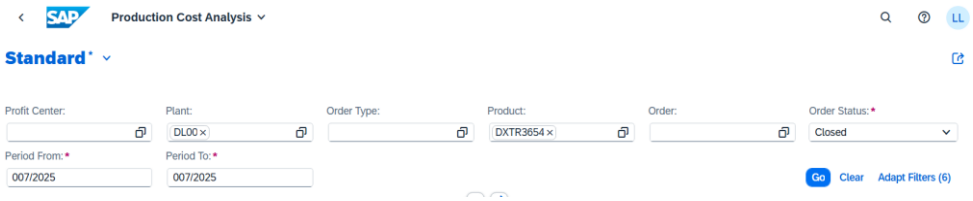
Start



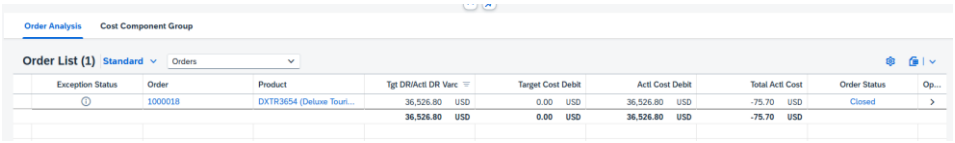
In the *Product* field enter **DXTR3###** and change the *Order Status* from **Open** to **Closed**

DXTR3###

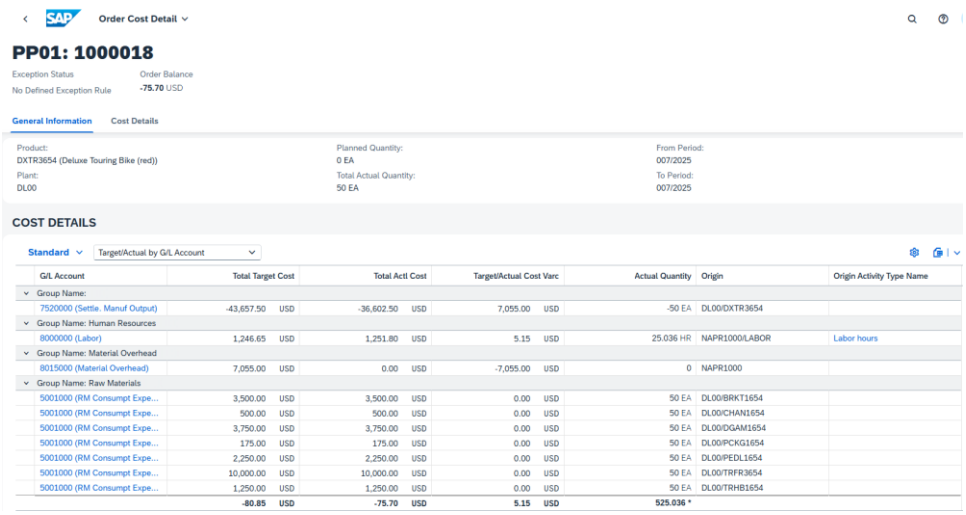
Closed



Click on Go to start the search. Your recently completed production order will be displayed.



This overview lists the summed target and actual costs and shows any variances. Click on › at the end of the entry to open the cost details.



Note Pay only marginal attention to overhead costs. In this case, they are shown in the target costs but not in the actual costs.

Now that the finished products have been received in the Inventory, the Manufacturing Output Settlement Variance has been added. How is this figure calculated by the system?

Click SAP to return to the SAP Fiori Launchpad.

Step 15: Settle Costs of Production Order

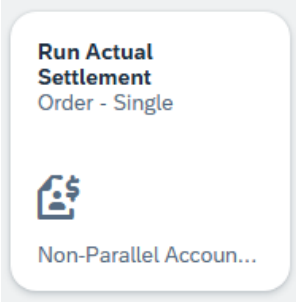
Task Settle costs of your production order. **Time** 20 min

Short Description Settle the costs of your production order. The costs are temporarily captured in the production order and they need to be assigned to an appropriate cost object. Compare the actual costs to the planned costs to identify any deviations or potential problems in this regard.

Name (Position) Jamie Shamblin (Controller)

To settle costs of a production order, in the space *Production Planning and Execution* space and the role of use the app *Run Actual Settlement - Order - Single*.

Start



If you have to input the Controlling Area, enter **NA00**, and click on Continue.

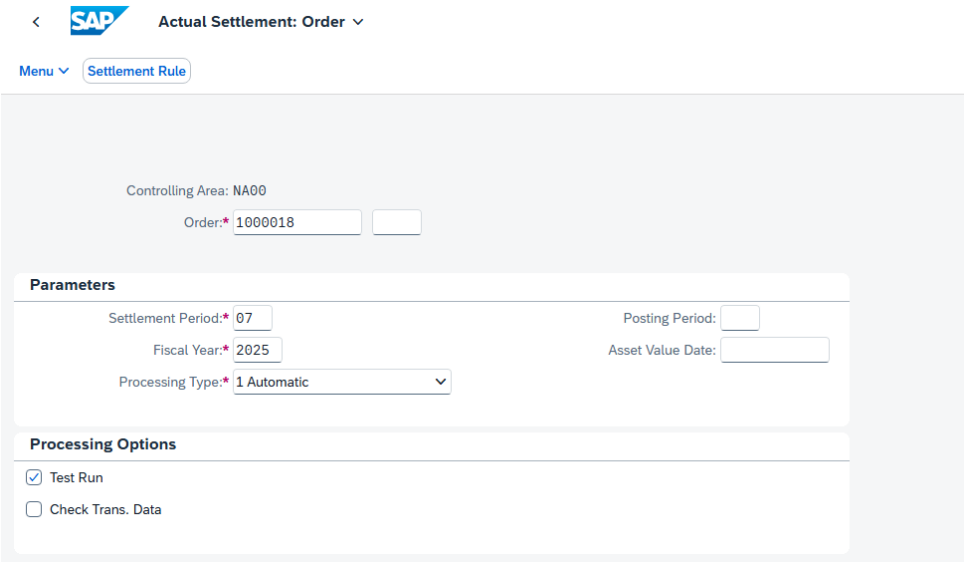
NA00

Enter your **production order number**, alternatively search for it with your material DXTR3### in the Input-Help. In the *Parameters* section enter *Settlement- and posting period* as **current month** (e.g., 006 for June). Enter as *Fiscal Year* the **current year**. Make Sure that **Test Run** is checked in the *Processing Options* section.

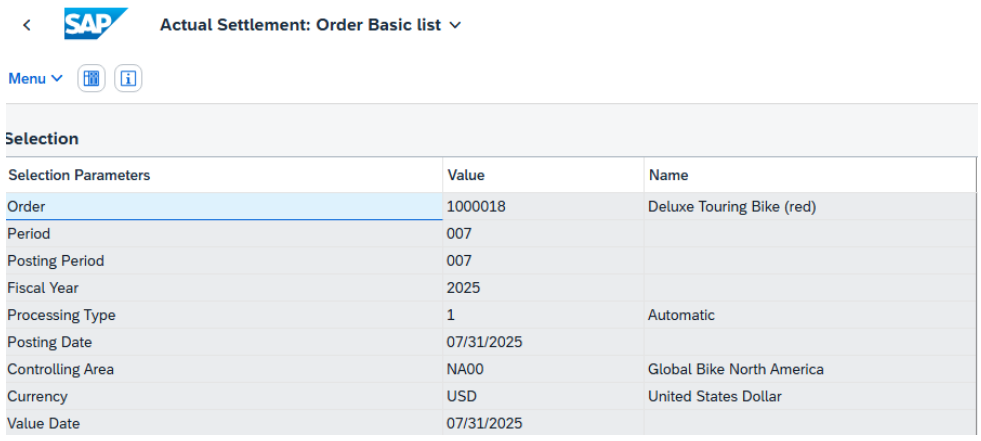
Order number

current month
current year

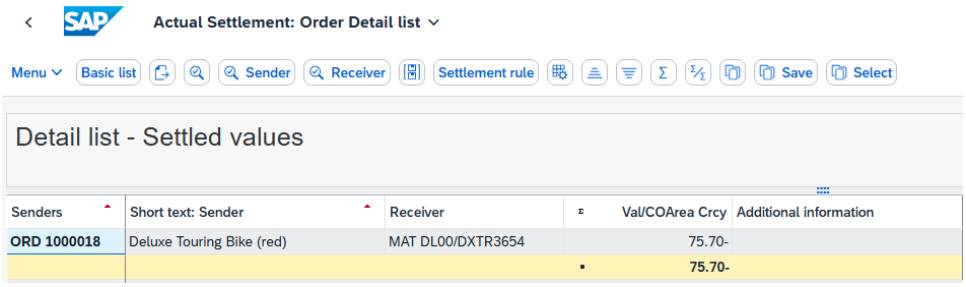
Test Run



Click on Execute to proceed. Confirm any occurring pop-ups with enter. You enter the screen *Actual Settlement: Oder Basic list*.



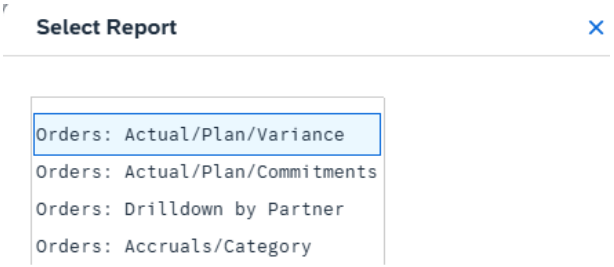
Click on in the upper part of the image to open the detailed lists.



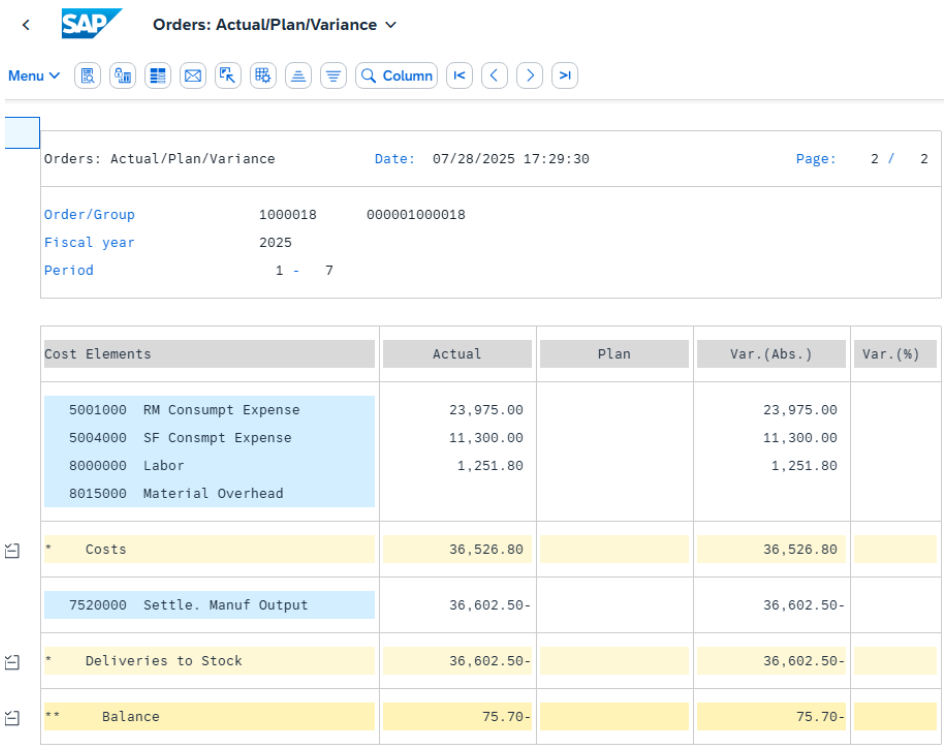
From the Menu bar choose: Menu Bar

Menu ► Environment ► Report

The *Select Report* pop-up will appear.



Click on **Actual/Plan/Variance** and confirm your selection with . A corresponding report group is generated and displayed. Actual/Plan/Variance

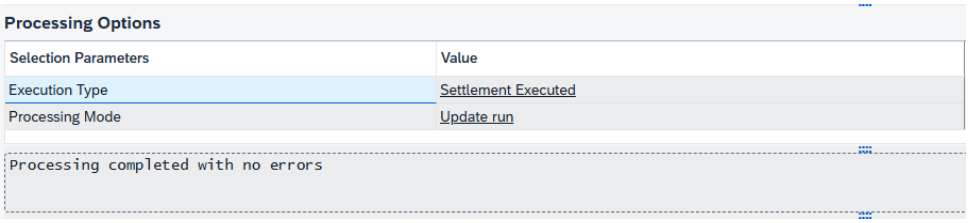


The test run is now complete and the *actual settlement* should now be performed.

Click on ‹ to go back. Confirm the Message to leave the report Yes .Afterwards press ‹ two times to get back to the entry screen.

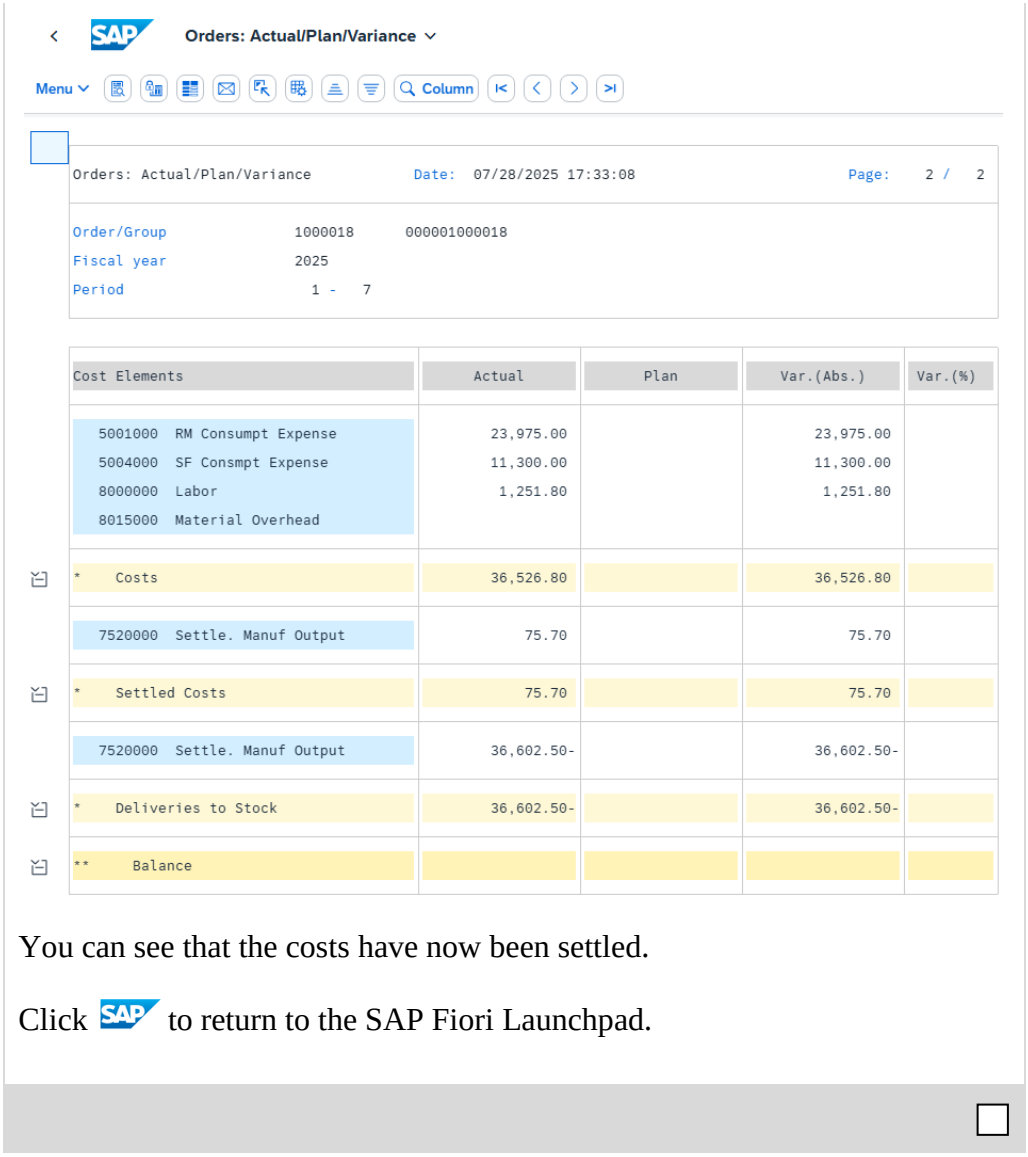
Deselect Test Run and again click on Execute . In contrast to the previous run, you can now see in the *Processing Options* area that this was an update run that was *completed with no errors*.

~~Test Run~~



Open the report **Actual/Plan/Variance** again by clicking on first and then use the menu bar path **Menu ► Environment ► Report** and select the option **Actual/Plan/Variance**.

Actual/Plan/Variance



You can see that the costs have now been settled.

Click SAP to return to the SAP Fiori Launchpad.

PP Challenge

Learning Objective Understand and perform an integrated manufacturing process. **Time** 60 min

Motivation After you have successfully worked through the *Production Planning and Execution* case study you should be able to solve the following challenge on your own.

Scenario In this challenge, you should plan the production for the product group Offroad Bicycles for the next months. Take into consideration that the materials of the product group have to be assigned to the strategy group (again "40"). You do not have to change the routing.

Therefore, enter manually the following planned independent requirements:

Period	**PIRs**
Current month + 1	110
Current month + 2	150
Current month + 3	175
Current month + 4	200
Current month + 5	85
Remaining months	100

The challenge focus will be on ORMN1###. For that product, please convert the first planned order into a production order. Then, carry out the production process. When posting goods receipts and issues, note the bill of materials for ORMN1###, which differs from the one in the case study. Once production is complete (including all goods movements), you settle costs afterwards.

Task Information Since this task is based on the *Production Planning and Execution* case study you can use it as guidance. However, it is recommended that you solve it without any help in order to test your acquired knowledge.

